Formalizing the IUT III 3.11–3.12 Corridor in Lean

IUT Lean formalization project

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Abstract

This paper reports the current state of a Lean formalization project for a narrow part of Mochizuki's inter-universal Teichmuller theory: the corridor from IUT III, Theorem 3.11 to Corollary 3.12. The project is deliberately neutral about the correctness of IUT as a proof of the *abc* conjecture and about the Mochizuki–Scholze–Stix dispute. Its purpose is to make the comparison levels, indeterminacy actions, real-line transports, and source obligations explicit enough that the proof assistant can distinguish proved consequences from mathematical hypotheses still to be constructed.

The formalization now verifies the ordered-real endpoint of Corollary 3.12, separates the representative j^2-level from full-label, averaged, and hull/log-volume comparison levels, and builds a typed route through concrete Hodge-theater/log-theta-style source data, $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$indeterminacy records, quotient possible images, a selected q-region, Remark 3.9.5-style hull/determinant data, and the existing C_Θ endpoint. The possible-image layer is now witness-bearing: class-level possible images, quotient pullbacks, and the selected q-region carry explicit nonemptiness data through the Step (xi) valuation-ball/log-shell source chain. The class/log-theta-lattice formula and typed fiber-transport sources now carry these witnesses as part of the same Theorem 3.11 spine; the top-level additive-Haar arithmetic-divisor audit consumes the corresponding nonemptiness proof to build the witness-bearing Theorem 3.11 possible-image record, rather than using a vacuous possible-image predicate. The arithmetic-divisor and record Ob3/Ob5 additive-Haar constructors can now obtain all-choice possible-image nonemptiness directly from the class-indexed Theorem 3.11 image law and the record/source image equality; the same witness bridge constructs the central Theorem 3.11 possible-image source-data record and its nonemptiness audit. The current Step (xi) milestone now also has a constructed theta-region completion audit: the preferred public route and the norm-square/Ob7 evidence are built from the same theta-region principal valuation-ball source, localized Step (xi) C_Θ source, and restriction-calibrated p-adic source. The Theorem 3.11 record boundary has also been lowered: a primitive constructor now derives the multiradial source record, the transport-base constructed qualitative/SHE bundle, and the constructed transport guards from explicit input-prime-strip, structured-SHE, algorithmic-parallel-transport, certificate-equality, Hodge-theater/log-theta/ log-Kummer, and target-region lattice source fields. A further packet layer now projects the unified source spine, target-region lattice formula, bridge-certificate constructor, and primitive constructor from those concrete fields. The packet also projects a vertical log-Kummer packet-aligned one-sided multiradial source, so its own theta-pilot images, gluing torsor, selected q-choice, and vertical log-Kummer source feed the finite F_ℓ procession/equality-quotient audit. The packet-route audit checks that the bridge certificate generates the primitive qualitative source and primitive constructor audit used by the public Step (xi) route. The qualitative side has also been linked to the Theorem 3.11 IPL/SHE/APT bridge package by an explicit source-copy and output-family alignment layer. The concrete bridge component can now be constructed from the lower constructed qualitative certificate source, and the packet audit exposes the corresponding no-collapse guards: forbidden direct SHE domain-to-codomain identification, permitted APT quotient transport, and selected IPL source/target matching. A further alignment source now packages the constructed qualitative certificate together with

the packet-specific bridge alignments and builds the primitive packet constructor from that source. The SHE/common-container alignment has been lowered one step further: it is now derived from constructed qualitative common-container data rather than supplied as a standalone packet equality. The input and selected output prime strips can now also be chosen canonically from the constructed qualitative IPL source. The remaining lower task is to construct this packet from the analytic multiradial, log-theta, local arithmetic, and IUT IV estimates of the papers. The current result is a checked conditional Stage 1 corridor, not a closed proof of Corollary 3.12 from Mochizuki's published hypotheses.

The current audit now freezes one canonical Stage 1 route instead of treating new endpoint variants as progress by themselves. Its remaining-assumption manifest now names the same-index q-normalized local-arithmetic-degree route and classifies each public input as constructed, derived, interface-only, or still a source obligation. On this canonical route the p-adic finite localized hull-vector-bundle source is derived from the finite-extension restriction-calibrated local arithmetic handoff, and the canonical C_Θ-scale equality is derived from the q-normalized theta/log-volume identity plus q-pilot positivity. The stronger case-bounded IUT IV residual source is now a constructor into the local arithmetic-degree residual package, not the canonical public residual input. The audit includes one-place diagnostics for the detached p-adic scale failure, the necessity of the signed $q \leq \theta$ comparison for $C_\Theta \geq -1$, and the remaining residual lower-bound obligation at the local arithmetic-degree boundary. Future Lean steps should remove further entries from the manifest, especially the weighted determinant synchronizations, q-normalized theta/log-volume identity, and local arithmetic-degree residual package.

We also maintain a Lean blueprint and intend to make the project consumable by Apodeixis as a collaboration and review layer. The blueprint is presently a compact navigational document plus a generated declaration index; it is useful for orientation, but it is not yet a complete mathematical dependency graph. This paper replaces an earlier ledger-style draft. Future updates should revise the relevant sections of this research paper rather than append chronological implementation notes.

1 Introduction

IUT is large enough that a useful formalization cannot begin by translating all four papers in order. The present project instead focuses on the public pressure point around IUT III, Theorem 3.11 and Corollary 3.12. Mochizuki's 2026 formalization report identifies this as the first stage of a possible formalization program:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The label "3.11.5" is not a theorem in the IUT III paper. It names the reorganized checkpoint obtained by moving the holomorphic hull plus determinant operation before the final simultaneous comparison of the Θ-pilot and the q-pilot.

The immediate formal target is not the *abc* conjecture. It is the logical shape of the Stage 1 implication. The question is:

> If the source constructions claimed by IUT III supply the multiradial possible images, indeterminacy behavior, IPL, SHE, APT, hull/determinant operations, and local-to-global C_Θ estimates, does the Corollary 3.12 inequality follow in a type-correct way?

The current code answers a substantial conditional version of this question. It proves the final ordered-real implication, constructs many intermediate typed routes, and exposes the remaining source mathematics as named obligations. It does not yet prove that the full Hodge-theater, Frobenioid, log-Kummer, holomorphic hull, determinant, and IUT IV analytic estimates of the source papers instantiate those obligations.

2 Mathematical Target

2.1 The final inequality

The endpoint formalized in Lean is the signed real comparison used in the Corollary 3.12 corridor. In the notation of the project, the route must supply

$$q_{\mathrm{signed}} \leq \Theta_{\mathrm{signed}}, \qquad 0 < -q_{\mathrm{signed}}, \qquad \Theta_{\mathrm{signed}} \leq C_\Theta(-q_{\mathrm{signed}}).$$

From these hypotheses the ordered-real calculation gives

$$-1 \leq C_\Theta.$$

The Lean theorem behind this step is elementary. This is important: the hard question is not whether the final real inequality follows from the displayed signed comparisons. The hard question is whether the source machinery really constructs the signed comparison without collapsing distinct histories or comparison levels.

2.2 The disputed comparison level

Scholze and Stix criticize the proof around Corollary 3.12 on the grounds that the relevant Θ-side data carries j^2-scaled information, while a naive simplification of the comparison appears to remove the room needed for that data. The formalization responds by enforcing separate types for:

representative j^2-coordinate data,
balanced sign-compatible data,
full-label and averaged finite-label data,
transported ordered-real copies,
hull/log-volume endpoint data.

Lean then rejects the most naive collapse: the balanced sign-compatible layer does not by itself supply the final Corollary 3.12 route. This is a diagnostic result, not a settlement of the dispute. The remaining issue is whether the source construction gives the final hull/log-volume object with the required compatibilities.

2.3 Indeterminacy behavior

The current Stage 1 interface treats the three indeterminacy components separately. In the formal model, Ind_1 and Ind_2 preserve procession-normalized log-volume and generate equality for the possible-image quotient. Ind_3 is not an equality generator; it contributes only an upper-semi log-volume inequality. This distinction is now part of the checked Lean interface:

$$\mathrm{Ind}_1, \mathrm{Ind}_2 : \text{equality/invariance}, \qquad \mathrm{Ind}_3 : \text{upper-semi inequality}.$$

This separation is one of the central design decisions of the formalization. It prevents a proof from turning an upper-semi source relation into an equality merely because both appear in the informal phrase “indeterminacy”.

3 Formalization Architecture

3.1 Source, Lean, paper

The working cycle is:

$$\text{Mochizuki source papers} \longrightarrow \text{Lean definitions and theorems} \longrightarrow \text{research paper and blueprint.}$$

The papers in `docs/` are treated as source of truth. The Lean code is the formal artifact. This paper records the mathematical state of the formal artifact; it is not meant to be a chronological log. This distinction matters because previous versions of this draft had grown into an implementation ledger. The present version is organized as a research paper: target, architecture, results, trust boundary, tooling, and next work.

3.2 Main Lean layers

The Stage 1 development is split into modules whose roles are stable enough to describe at paper level:

Module family	Role
`PilotComparison`, `CorollarySchema`, `SourceObligations`	The signed real endpoint, the final Corollary 3.12 predicate shape, and the ledger turning a common-bound comparison into the endpoint.
`IUTStage1SourceCore`, `IUTStage1FiniteLabels`, `IUTStage1Gaussian`	Real-line copies, regions, finite-label and $ZMod(\ell)$ models, Gaussian degree shadows, and j^2-diagnostic infrastructure.
`IUTStage1StepX`, `IUTStage1HodgeSHE`	Procession-normalized log-volume routes, Hodge/SHE data, square/full-label transport, and upper-semi compatibility records.
`IUTStage1Theorem311`, `IUTStage1StepXI`	Typed Theorem 3.11 indeterminacy cores, possible-image quotient structures, concrete Hodge-theater/log-theta source layers, Remark 3.9.5 hull/determinant interfaces, and paper-trace audits. The public `IUTStage1StepXI` module is now a facade over a core implementation module, preparing the Step (xi) layer for canonical-route source slices without changing public names.
`IUTStage1FrobenioidShift`, `IUTStage1Experiments`	Nonarchimedean realified Frobenioid/log-Kummer and vertical-IQ routes, additive-Haar arithmetic-degree endpoints, and compiled public diagnostics.

The names in this table are intentionally coarse. A paper should not reproduce thousands of declaration names. The detailed declaration surface is maintained in the blueprint's generated declaration index.

3.3 Records as boundaries

The project uses records in two different ways. Some records are ordinary mathematical data structures whose fields are proved or constructed in Lean. Other records are trust-boundary interfaces:

they name a source construction that must eventually be derived from the IUT papers. The current paper distinguishes these roles explicitly.

For example, the typed indeterminacy core is a useful formal object: it stores the $\mathrm{Ind}_1/\mathrm{Ind}_2$ invariance laws, the Ind_3 upper-semi law, and the quotient relation. By contrast, the full Theorem 3.11-and-remarks obligation record still contains source-facing fields such as construction of the input prime-strip link, theta-pilot possible images, the log-theta-lattice procession, and the Hodge/SHE/IPL/APT bridge. These fields are not hidden axioms, but they are also not yet source-derived mathematics.

4 Current Formal Results

4.1 Ordered-real endpoint

The final ordered-real implication is formalized. Once the signed comparison $q_{\text{signed}} \leq \Theta_{\text{signed}}$, positivity of $-q_{\text{signed}}$, and the C_Θ-upper bound are available, Lean derives the expected lower bound $-1 \leq C_\Theta$. The code also proves the corresponding strict/boundary dichotomy: either the boundary case is exact, or $-1 < C_\Theta$.

This part is no longer an open mathematical problem inside the project. It is pure ordered-real algebra. All serious mathematical work sits upstream of the signed comparison and the C_Θ-upper bound.

4.2 Anti-collapse diagnostics

The formalization proves negative diagnostic statements about comparison-level collapse. In particular, a label-independent j^2 collapse is rejected at the representative level, and the balanced sign-compatible layer is not accepted as the final route level. These diagnostics are valuable because they test the exact failure mode emphasized in external criticism.

The diagnostics should be read carefully. They show that the Lean development does not silently make the criticized simplification. They do not show that Mochizuki's construction successfully supplies the stronger hull/log-volume data.

4.3 Concrete Stage 1 route

The current strongest route starts with a unified Hodge-theater/log-theta/log- Kummer source spine. This spine carries the 0- and 1-column histories, prime-strip data, theta-pilot possible images, the selected q-choice, and the vertical log-Kummer packet alignment. Its Theorem 3.11 bridge side is also structured: the input-prime-strip link, one-column simultaneous holomorphic expression, and algorithmic parallel transport are projections of one IPL/SHE/APT bridge package. The Step (x) side is represented by a single finite-divisor/vertical-log-Kummer package: finite divisor tensor packets, realified Frobenioid Kummer compatibility, mono-analytic forgetting, vertical IQ, and source-target log-volume calibration are projections of one object rather than five unrelated source flags. Lean projects from the same spine both the Theorem 3.11 typed $\mathrm{Ind}_1/\mathrm{Ind}_2/\mathrm{Ind}_3$ core and the Step (x) payload. The route then forms the equality-quotient family of possible images, selects the q-region, and connects that region to the existing C_Θ/Corollary 3.12 corridor. The preferred paper-trace route now has a named source-spine audit: the Theorem 3.11/Remarks obligation package and the Step (x) finite-divisor/log-Kummer obligation package are projected from one source object. The first Step (xi) hard-math route has also been added: a paper-derived Remark 3.9.5/Step (xi) source records the substeps (xi-a)–(xi-g), ties the selected q-region to the actual Theorem 3.11 possible image selected by the source spine, and projects the existing Step (xi) hull/determinant obligation record internally. Ob7 is no longer only a raw compatibility flag on the newest route: the legacy prime-strip/log-Kummer record and its input/output strip and selected-column equalities are projected from a coric $\Theta^{\times\mu}$/log-Kummer source link aligned with the same quotient hull source. The preferred localized route now exposes this source as a structured audit: q-region containment, Gaussian/environment place agreement, coric unit-character agreement, and the three hull-bridge alignments are checked before the legacy compatibility record is projected. The constructed-paper-trace public boundary now also packages this information together with the constructed Step (xi-a)–(xi-g) and Ob1–Ob9 ledger, making explicit that Ob7–Ob9 are read from the same log-Kummer/bridge source rather than supplied as separate trace facts. Its strictest boundary now derives the non-Step-(xi) audit from the proof-carrying Theorem 3.11, Step (x), and IUT IV source records; the caller supplies only the explicit additive-Haar arithmetic-degree payload that remains below this milestone. The selected q-region containment is now derived from the equality-quotient possible-image family: it is the canonical inclusion of the selected possible region into the family union, followed by the constructed holomorphic hull source. The possible-image family is no longer allowed to be an empty predicate shell at this boundary: the theta-pilot class image source carries a point witness for every class, Lean pulls these witnesses back to concrete choices, and the equality-quotient construction proves nonemptiness of the quotient pullbacks and of the selected q-region. The Step (xi) valuation-ball/log-shell source chain now transports the same theta-region witnesses down to the local exact-source constructors. More recently, the class-region, class-formula, log-theta-lattice formula, lattice-image-law, and typed fiber-transport source records themselves were made witness-bearing, so this nonemptiness data is no longer introduced independently by the valuation wrapper. The paper-derived Step (xi) source also has a quotient-compatible constructor that fixes the selected choice to the source spine, instead of accepting an independently assembled hull record at that point. The remaining IUT IV and additive-Haar layers are then combined with this constructed Step (xi) source. The concrete-packet goal-facing completion audit

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preferredPublicConcreteStepXI311312ConcretePacketTheorem311ConstructedThetaRegionGoalCompletionA
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packages the current milestone into two checked claims built from the same packet-derived primitive constructor: the constructed theta-region preferred public route and the constructed theta-region current product-hull norm-square/Ob7 evidence. Its scope is:

> theta-region principal valuation-ball source,
> concrete Theorem 3.11 packet projecting the primitive constructor,
> theta-pilot-class target-region formula with point witnesses,
> component-level Step (x) finite-divisor/log-Kummer source,
> bridge-derived IPL/SHE/APT witnesses with structured output-flag calibration,
> principal-product HDD Step (xi) source built by Lean,
> localized finite-place C_Θ handoff,
> restriction-calibrated p-adic source and Ob7 norm-square projection.

The audit is not a proof of the lower analytic source constructions. Its function is to ensure that, at this milestone boundary, the public route no longer combines an arbitrary constructor-built Step (xi) source with unrelated Ob7 evidence or a standalone hull/determinant obligation package. The new primitive Theorem 3.11 constructor further prevents the source record, constructed-SHE bundle, and transport guard from being independent top-level inputs: the primitive endpoint reconstructs them from the source spine and the lattice-formula realization of target regions, then enters the localized Step (xi) layer through a primitive principal-product HDD source whose audit records the induced source gap and transport guard. The concrete packet constructs that primitive endpoint through a bridge-certificate constructor, so the primitive qualitative source is obtained from the transported IPL/SHE/APT bridge package rather than supplied beside it. The packet now also has an output-flag-calibrated source-spine constructor: the deepest component-representative selected-label q-pilot route constructs the Theorem 3.11 subclaims from the package-aligned bridge and the single structured output-flag equality, instead of accepting an independent subclaim component. The same lowering now reaches the one-sided source-spine-data constructor used by the multiradial packet route, so that lower one-sided path also consumes the structured output-flag calibration rather than a standalone subclaim component.

4.4 Local-to-global C_Θ chain

The code now contains a proof-carrying local-to-global C_Θ route at the interface level. The preferred additive-Haar arithmetic-degree and p-adic formula route threads local analytic estimates, arithmetic-degree calibration, finite-place Haar defects, formula-gap matching, and Step (xi) local terms into the endpoint. For this preferred route, the canonical C_Θ-scale comparison is derived from the finite-place arithmetic gap and the IUT IV handoff identity, rather than supplied as a raw endpoint hypothesis. At the Step (xi) source layer, Lean now also proves the sharper handoff for both the constructed Remark 3.9.5 hull/determinant source and the constructor-built exact-theta source: once the local canonical Step (xi) scale is identified with the finite-place scale in the IUT IV summation record, the global C_Θ-dominance follows from the IUT IV plus-one cancellation theorem. The constructor-built finite-place source lowers this boundary again. It records local canonical scales, local arithmetic and main-log contributions, and local Haar normalization defects; Lean sums the local Step (xi) inequalities and uses the total defect bound to derive the global arithmetic gap consumed by the coupled IUT IV C_Θ ledger. The strongest concrete packet route now exposes the resulting signed inequality $\theta_{\text{signed}} \le C_\Theta(-q_{\text{signed}})$ and the Corollary 3.12 dichotomy as endpoint fields, not only as an internal localized-handoff audit. A further wrapper removes the independent principal product hull argument from this path by taking it from the packet-backed theta-region source. The next wrapper constructs that theta-region source from the principal valuation-ball product-hull cover and the fiber-transport family-union comparison attached to the concrete Theorem 3.11 packet. The current strongest variant lowers the boundary once more: it accepts a single theta-region-defined principal valuation-ball source and projects both the principal cover and fiber-backed family-union comparison internally. A pointwise metric variant now goes below that source by accepting the principal constructed-log-shell/metric-zero valuation-ball data and projecting the theta-region-defined source from it. The pointwise route also projects the theta-region measure/summand calibration from explicit Hodge/measure formulas: the hull-volume identity, the Hodge-Arakelov theta-degree charting identity, and the finite adjusted-summand formula. Thus the packaged pointwise calibration record is no longer a public endpoint object. The same route now constructs the restriction-calibrated norm-square Ob7 handoff from a pointwise residue-module/unit-ball/additive-Haar local arithmetic source. The local-analytic Theorem 1.10 refinement now keeps this proof-carrying pointwise Ob7 source as the public local-arithmetic boundary, so the structure-sheaf/direct-summand and nonzero norm-square projection facts are internal fields of that source rather than separate endpoint arguments. The newer product-handoff refinement constructs the packet-facing pointwise Ob7 source internally from the principal-product restriction-calibrated handoff, so the local-analytic Hodge-formula route can expose that lower product handoff instead of the pointwise wrapper. This branch is now explicitly quarantined: the product-level and packet-facing handoff contradiction theorems show that the unit-ball norm-square source carries both the selected zero-norm computation and the selected nonzero-norm field. A constructor route from the lower restriction-calibrated residue-module/unit- ball/additive-Haar source remains available; at that lower boundary those two projection facts are still explicit local arithmetic bridge fields. A new parallel pointwise branch now avoids that quarantined unit-ball positive-norm input: it couples the concrete pointwise theta-region principal hull to the residue-module inverse-base-prime/additive-Haar source and derives the localized C_Θ signed comparison from that branch together with the localized Step (xi) handoff. A localized Step (xi) refinement now also packages the localized determinant source, its determinant and canonical-scale identifications, local main-log terms, Haar defects, and finite-place summation identities into a single pointwise source object before the public endpoint projects them. The IUT-IV-backed refinement lowers this finite-place layer further by deriving the arithmetic-upper and main-log summation identities from the IUT IV arithmetic-divisor local evaluation source, leaving only the local Step-(xi)-plus-Haar identification as the visible comparison datum. The latest Theorem 1.10 valuation-ball refinement lowers the IUT IV input one more step: the arithmetic-divisor local evaluation source is now projected from the valuation-ball additive-Haar formula-gap source, whose endpoint contains the distinguished Proposition 1.4 log-shell construction, nondistinguished zero case, archimedean Proposition 1.5 metric construction, and named formula-to-gap comparisons. The corresponding inverse-base-prime pointwise route now carries this same lowered Hodge-formula and Theorem 1.10 valuation-ball/IUT IV interface, while using the residue-module inverse-base-prime additive-Haar source instead of the quarantined unit-ball norm-square local arithmetic handoff. This branch has also been lowered one step further: the pointwise route can now construct the restriction/additive-Haar wrapper from finite Hodge-summand/Gaussian-Hodge projected inverse-base-prime data. The Theorem 1.10 valuation-ball side has likewise been lowered: the strongest pointwise branch now constructs the formula-gap matched valuation-ball source from local analytic arithmetic-divisor data before entering the localized Step (xi) handoff, and the local-analytic and Theorem 1.10 C_Θ paper-trace records are now projected from that valuation-ball additive-Haar source instead of supplied as parallel records. This projected source is now threaded into a public pointwise route, so the assembled local-analytic Step (xi) C_Θ wrapper is not a separate endpoint input there. The heterogeneous projected-weighted determinant route now exposes both the signed dichotomy and the $C_\Theta \geq -1$ companion directly from the q-normalized local-analytic residual source, before passing through the formula-gap wrapper. The same lowering now applies on the non-vacuous inverse-base-prime/additive-Haar pointwise branch before the summand-projected signed comparison is invoked, and the compact-open log-Kummer route propagates this constructed wrapper through the possible-region source. It also now constructs the principal pointwise theta-region valuation-ball source from compact-open log-Kummer map possible-region data at the public endpoint. The newest compact-open branch lowers the inverse-base-prime arithmetic input one step further: the public endpoint takes the residue-module valuation cover and finite Hodge-summand Gaussian/Hodge construction separately, and Lean assembles the summand-projected inverse-base-prime source internally. A companion branch also constructs the local-analytic Step (xi) C_Θ wrapper at this valuation-cover/Hodge-summand boundary from the localized determinant source, Haar defect, and valuation-ball additive-Haar local analytic arithmetic-divisor data. A newer variant lowers the Hodge side again: the endpoint takes the localized-adjusted Gaussian/Hodge source, and Lean reconstructs the finite Hodge-summand source by fixing the summands to the localized weighted-adjusted determinant contributions; its constructed-wrapper companion performs the same local-analytic Step (xi) C_Θ construction at this lower localized-adjusted boundary. A further square-direct-summand variant replaces the direct nonnegativity and strict-positivity fields with square-root presentations of the local direct-summand log-volumes and a nonzero selected root, and its constructed-wrapper companion moves the same local-analytic Step (xi) C_Θ construction below these square-root data. The norm-square variant lowers this again by projecting those square-root presentations from the norm-square localized hull decomposition, and its constructed-wrapper companion moves the local-analytic Step (xi) C_Θ construction below that norm-square projection. The compact-open norm-square variant moves the public Hodge datum down to local compact-open tensor factors whose projected log-volumes generate the norm-square decomposition; its constructed-wrapper companion moves the local-analytic Step (xi) C_Θ construction below those compact-open tensor factors. The additive-Haar variant lowers the same branch to Haar-normalized compact-open local factors and projects the compact-open norm-square source from their normalized log-volume identities; its constructed-wrapper companion now performs the local-analytic Step (xi) C_Θ construction at this lowest compact-open Hodge boundary. Its pointwise-source strengthening now reads the localized determinant/scale synchronization, formula-gap Theorem 1.10 valuation-ball source, Haar defect, local Step (xi)/Haar/main-log identity, and finite Haar lower bound from a single localized Theorem 1.10/IUT IV Step (xi) source. The restriction-calibrated local arithmetic branch now has the same local-analytic Theorem 1.10 valuation-ball/IUT IV lowering as the inverse-base-prime branch. Its compact-open wrapper derives the pointwise valuation-ball source from log-Kummer map data and now also has a pointwise formula-gap variant that reconstructs the local-analytic wrapper from a single localized Theorem 1.10/IUT IV source. At the formula-realized evidence level, the same compact-open restriction-calibrated boundary now constructs the pointwise valuation-ball source and Ob7 local-arithmetic handoff before applying the pointwise Theorem 1.10/IUT IV packet route; its product-handoff variant now takes the typed Ob7 handoff without separately exposing the restriction-calibrated source, structure-sheaf/direct-summand equality, or selected norm nonzero field. A further local-arithmetic-degree variant now constructs the pointwise Theorem 1.10/IUT IV localized source from the localized local-term Step (xi) source, formula-gap valuation-ball source, main-log matching, and local arithmetic-degree identity. The synchronized variant removes that localized local-term source from the compact-open public boundary by constructing it from determinant synchronization, Haar defects, and the finite-place arithmetic-degree identity. The constructed-synchronization variant also removes the packaged determinant/scale synchronization source by constructing it from the localized Step (xi) determinant source and the two constructor-built comparison identities. The explicit-formula variant removes the packaged pointwise measure/summand formula record by constructing it from the volume/hull-log-volume, charted-theta/Hodge-degree, and finite adjusted-summand identities. The realized-exact compact-open variant also constructs the compact-open log-Kummer map source from the statement that each theta-region cell is the realized image of the selected compact-open local subset. Its restriction-calibrated variant constructs the product Ob7 handoff from the normalized local-arithmetic source, structure-sheaf/direct-summand synchronization, and nonzero norm-square summand.

The phrase "proof-carrying" is deliberate. The route no longer consumes a single unstructured numerical inequality at the public endpoint. It consumes a structured chain of local and global source objects. Nevertheless, several of those source objects still stand for arithmetic constructions not yet derived from number-field valuations, conductors, differents, holomorphic hulls, and the local estimates of IUT IV.

5 What Remains Mathematically Open

5.1 Theorem 3.11 from source data

The largest remaining mathematical task is to construct the Theorem 3.11 multiradial output from actual IUT I–III source data. The present code has a unified Hodge-theater/log-theta/log-Kummer source spine that projects to the typed core, possible-image quotient, input-prime-strip link, SHE/IPL/APT bridge, and Step (x) finite-divisor/log-Kummer payload. The preferred paper-trace API no longer asks for those Theorem 3.11 and Step (x) pieces as parallel flags; they are read from the spine. The implementation does not yet derive that spine, or the structured bridge and Step (x) packages inside it, from the full Hodge-theater, local LGP, and Frobenioid constructions. The current bridge component is no longer a fully free qualitative packet: Lean can construct it from the lower constructed qualitative certificate and audits the SHE/APT/IPL no-collapse projections at the primitive packet boundary. The packet has also gained a constructed-qualitative bridge-alignment source. The SHE/common-container alignment is now a projection from constructed qualitative common-container data, and the input/selected-output prime strips are canonical projections from the constructed qualitative IPL source. The one-column Hodge theater can now be fixed as the constructed SHE codomain, with the one-column history read from the selected q-choice. A still lower q-pilot-canonical bridge source now constructs the selected concrete q-choice with the constructed SHE q-pilot theater; Lean then derives its codomain placement from the structured SHE q-pilot/codomain law. A q-pilot-class bridge source goes below the concrete-coordinate boundary by rebuilding the selected q-choice from a theta-pilot lattice class and representative data for the forgotten finite label, procession state, local tensor state, and upper-semi state. A component-representative bridge source now lowers that representative section to the generated log-theta/vertical-log-Kummer/Frobenioid divisor-column component kernel. A source-spine q-pilot component bridge now reads the q-pilot history label, label-forgotten log-theta lattice coordinate, and coric datum from the selected q-choice of the unified Theorem 3.11 source spine. A generated-full-label source-spine bridge lowers the last equality further by matching the source-spine selected q-choice to the concrete projection of the generated full-label representative over the q-pilot theta class; Lean derives the component-kernel representative equality from the full-label kernel theorem. A generated-q-choice source-spine bridge now lowers this once more: it supplies the generated q-choice itself, proves that its theta class is the q-pilot theta class and that its finite label is the component kernel's representative label, and derives the generated-full-label representative projection. A selected-label source-spine bridge removes that generated q-choice as a public object: Lean constructs it internally as the kernel representative full-label choice over the q-pilot theta class read from the source spine, proves that its finite label is the selected source label, and derives the selected q-choice projection from fieldwise source equalities for the q-pilot theater, representative finite label, procession state, local tensor state, and upper-semi state. A component-representative selected-label bridge lowers those fieldwise equalities to the vertical-log-Kummer/Frobenioid component representative source: applying its constructed-representative theorem to the selected q-choice and transporting across the q-pilot theater alignment yields the representative finite label, procession state, local tensor state, and upper-semi state internally. A one-sided component selected-label bridge now removes the independent component-representative source at this boundary: Lean projects it as the `volumeSource` of the concrete one-sided multiradial component package, and retains only synchronization of the one-sided selected q-choice with the source spine and constructed q-pilot theater. Its SHE-codomain successor derives that q-pilot theater placement from the source-spine one-column law, the constructed SHE codomain/one-column alignment, and the structured SHE q-pilot/codomain equality. A source-spine-selected successor now constructs the one-sided component source with the source-spine selected q-choice, making that synchronization definitional. A source-spine-data successor also projects the theta possible-image source and gluing torsor from the unified source spine. The primitive-packet successor now projects the multiradial record and target-region-calibrated source spine from the same primitive packet, so the record/source image-family agreement and the SHE codomain/one-column alignment are inherited from that packet rather than supplied as independent bridge inputs. A component-kernel successor now constructs the one-sided component volume source from the Frobenioid divisor-column representative kernel. The generated full-label package layer then avoids a global ambient-coverage assertion: Lean pulls the concrete pre-ledger and source package back along the kernel projection from generated full-label choices, provided only a selected generated q-choice whose projection is the pre-ledger chosen output. A selected-label package-reindex alignment now factors this through the source spine: Lean constructs the selected generated q-choice from the selected-label q-pilot bridge and derives its chosen-output projection from the single equality identifying the source-spine selected q-choice with the pre-ledger chosen output. A primitive-packet selected-label reindex alignment lowers this once more: the selected-label source is projected from the primitive packet's component-kernel one-sided source, and the selected q-choice of that projected source spine is definitionally the primitive packet's selected q-choice. A chosen-output primitive packet reindex alignment removes that equality as an external field by constructing the primitive packet with the pre-ledger chosen output itself. The chosen-output generated full-label quotient corridor then projects the generated kernel, theta-class image family, gluing torsor, and selected generated q-choice from the same chosen-output packet route. A constructed generated-corridor source now rebuilds the generated multiradial record with the full-label image family forced by that packet, so the image-family equality is no longer an external field. The next bridge boundary is therefore the chosen-output one-column/history placement, the bridge/component data indexed by that chosen output, and a generated SHE-structured Theorem 3.11 bundle: the base generated multiradial record is now constructed from this bundle before the forced image family is installed. This generated bundle is now itself obtained by reindexing the concrete structured bundle along the chosen-output full-label projection. At the direct product-hull/HDD surface, the chosen-output generated full-label wrapper now consumes this chosen-output corridor and projects the older generated-HDD source internally, while its reindexed-structured strengthening lowers the public route through this concrete-to-generated bundle constructor. This replaces the earlier public boundary of prebuilt q-pilot class fields, a prebuilt bridge component, an independent one-column/codomain equality, a supplied representative-data section, a direct constructed-representative equality, a direct generated-full-label projection, a source-provided generated q-choice, fieldwise selected-label compatibilities, an independent component representative source, a selected-q synchronization equality, an independent theta possible-image source, an independent gluing torsor, an independent record/source image-family synchronization, a monolithic one-sided component volume source, a raw q-pilot theater placement field, or an assertion that every ambient concrete choice is a kernel representative. The missing source construction includes:

initial theta data and Hodge-theater histories,
prime strips and the derivation of the structured input link,
the multiradial algorithm and its possible images,
the log-theta-lattice procession and F_ℓ-symmetry,
the derivation of the 1-column SHE/IPL/APT bridge,
the derivation of the Step (x) package from vertical log-Kummer data,
the proof that $\mathrm{Ind}_1/\mathrm{Ind}_2$ preserve normalized log-volume,
the proof that Ind_3 has only the stated upper-semi orientation.

The Lean records now say exactly what must be supplied. They do not yet replace Mochizuki's proof of Theorem 3.11.

5.2 Remark 3.9.5 and Step (xi)

The second major gap is the genuine holomorphic hull and determinant construction. The current code now has a paper-traced Step (xi) source route: the old Step (xi) obligation record can be projected from a structured source that records Corollary 3.12 Step (xi-a)–(xi-g), Remark 3.9.5 Ob1–Ob9, the selected Theorem 3.11 possible image, determinant data, and Ob7 link pinning. At the hull-formation boundary, the selected q-region is now constructed from the equality-quotient possible-image family, so its containment in the possible-image union is no longer a separately supplied Step (xi) field; the paper-derived source constructor uses the source spine's selected q-choice directly. This boundary has now been lowered one step further: the Step (xi) hull source may be built from the Remark 3.9.5 minimal-hull system $\phi(P) = \bigcap_{P \subseteq H} H$. Lean derives the holomorphic hull operator, the canonical hull of the possible-image union, closedness of this canonical hull, containment of the selected q-region in the union, and the Ob1/Ob2 absorption statement from this minimal-hull source rather than from a free hull-operator witness. The current lower constructor also accepts the principal product presentation $\lambda \cdot O$ of Remark 3.9.5(i): the base integer/valuation-ball region O, nonzero scalar parameters, scalar-image principal hulls, the intersection-parameter hull, and the principal-hull log-volume are projected into the concrete Step (xi) hull source. At the determinant boundary, the Step (xi) determinant comparison record can now be constructed from the weighted arithmetic-vector-bundle determinant source already present in the Step (xi) layer: the finite adjusted summands, determinant sum, normalization, and positive tensor power are projected from that lower source rather than supplied as independent fields. This has now been lowered to the localized arithmetic-vector-bundle decomposition source: the localized hull cover supplies a finite sum of localized hull-region log-volumes, each localized log-volume is identified with the corresponding structure-sheaf-adjusted vector-bundle contribution, and Lean projects the result to the Step (xi) determinant comparison data. The selected q-region-to-normalized-determinant inequality is then derived from hull containment plus the localized family-hull/normalized-determinant equality; the localized projection audit now also exposes the adjusted Ob3/Ob5 endpoint, the induced Ob3/Ob5 compatibility source, and the bounded-family hull/determinant log-volume endpoint proved in the lower SourceCore layer. This localized decomposition now also constructs the record-pinned Ob3/Ob5 adjusted determinant source itself, after an explicit identification of its possible-region family with the multiradial record family. Consequently the obligations-backed exact-theta possible-image Step (xi) constructor can be entered directly from localized hull/vector-bundle data, with the finite localized sum providing determinant compatibility and the obligations package providing the bridge, q-positivity, and normalization. The strict localized route now packages this with the equality-quotient Ob5 statement: choices in the $(\mathrm{Ind1}, \mathrm{Ind2})$ equality orbit have identical quotient possible regions and quotient-region log-volumes, while the same localized family-hull construction supplies the determinant and bounded-family tensor endpoint. The record Ob3/Ob5 boundary has also been lowered through the theta-region-defined principal valuation-ball source to the typed fiber-transport valuation source. Lean pulls the theta-pilot valuation cover back to concrete choices, proves that its adjusted possible-region family is the multiradial record possible-image family, builds the record Ob3/Ob5 adjusted determinant/log-volume source from this cover, and then feeds the resulting source into the exact-theta constructed hull/determinant API. For a lattice-image-law-backed source the remaining explicit input is the identification of the supplied lattice-image law with the concrete fiber-transport law; for the fiber-transport-backed source this equality is internal to the constructor. This boundary has also been lowered through the local log-shell/log-Kummer layer: a principal pointwise constructed log-shell metric-zero source now constructs the same record Ob3/Ob5 source, and compact-open log-Kummer image, correspondence, and map possible-region sources construct both the record Ob3/Ob5 source and the exact-theta Step (xi) hull/determinant source. Lean passes through the local log-Kummer graph or relation-level image laws, compact-open representatives, and the converse realization law before entering the principal valuation-ball constructor. The compact-open realized-exact source now also exposes the literal image equality $\Theta_v = \mathrm{realization}(U_v)$ and constructs the log-Kummer map source directly from it; the same realized-exact source now also constructs the Record Ob3/Ob5 adjusted determinant/log-volume source and the exact-theta Step (xi) hull/determinant source through that map layer. The Hodge-calibrated compact-open route has been lowered to this same realized-exact boundary: realized exactness constructs the Hodge-calibrated Step (xi) source, its side-conditioned form, and its source-hull-det-data-backed form. The exact possible-region, log-Kummer map, image, and correspondence endpoints now also have a Record Ob3/Ob5 valuation-ball source form: the Ob3/Ob4-to-Step (xi) determinant equality is projected from the record-canonical source rather than passed as a separate endpoint hypothesis. The corresponding choice-indexed arithmetic-formula routes at the exact possible-region, log-Kummer map, image, and correspondence boundaries now also construct the raw determinant compatibility from this Record Ob3/Ob5 source; their remaining synchronization input is the lower equality identifying the adjusted valuation-ball hull operator with the principal product hull. These four choice-indexed compact-open boundaries now also have constructor-sourced bridge variants at both the arithmetic-formula and formula-gap endpoints: the package hull/determinant bridge equality is projected from a Theorem 3.11 constructor-sourced Step (xi) source, while the remaining lower synchronization identifies the constructor bridge datum with the corresponding Record Ob3/Ob5 principal-product hull/tensor-power construction. The same constructor-sourced bridge form now reaches the choice-indexed arithmetic-degree-controlled, direct p-adic unit-ball Haar, and derived p-adic unit-ball Haar compact-open map, exact possible-region, log-Kummer image, and log-Kummer correspondence endpoints: the lower comparison, local Haar, and derived formula-matching data still supply the formula-gap source and the determinant/C_Θ-sum identifications, but the hull compatibility and package bridge are reconstructed from the Record Ob3/Ob5 source and constructor bridge. The correspondence route now also has an obligations-backed entry point: the package bridge equality, q-pilot positivity, and normalization are projected from 'IUTStage1SourceHullDetObligations'. Its Hodge-calibrated variant removes the raw theta/family-hull scalar equality from this boundary: Lean derives Θ_{signed} = family-hull log-volume from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', via the Hodge–Arakelov theta monoid degree. There is also a direct bridge-based compact-open log-Kummer map entry point for this Hodge-calibrated route: the realization map and local sections construct the correspondence relation internally, and the constructor uses the Hodge-derived theta/family-hull equality without an ambient 'IUTStage1SourceHullDetObligations' object. A side-conditioned form of this map route now projects q-positivity and normalization from 'IUTStage1SourceSideConditions'. A further source-data-backed form projects the package hull/determinant bridge equality from 'IUTStage1SourceHullDetData', leaving only the synchronization of that stored bridge datum with the record-canonical hull/tensor-power construction, together with measure calibration, as explicit lower input. The older obligations-backed map entry point is retained as a compatibility projection. The same calibration now reaches the finite-divisor constructor-backed source: Lean converts the Hodge-calibrated data to the constructor-backed spine, forms the constructor-generated hull/determinant obligations, and builds the possible-image Step (xi) source without reintroducing a raw Θ_{signed}-to-family-hull equality at this boundary. The target-charted exact-theta adjusted-summand source now enters this Hodge-calibrated finite-divisor constructor first and projects the older constructor-backed source only for legacy consumers. The all-in-one target-charted Hodge/IPL possible-image route audit also exposes the record-canonical Theorem 3.11 hull/tensor-power bridge equality beside the ambient-obligations bridge equality, making this synchronization visible at the public route boundary instead of only through the obligations projection. It now also carries the target-charted reduction to the constructor-built Step (xi) source and the resulting Remark 3.9.5 Ob1–Ob5 bridge audit, so the public route records the actual possible-image hull absorption and determinant-compatibility chain below the canonical bridge equality. At the constructor-built boundary itself, Lean now converts the constructor-built possible-image Step (xi) source into the constructed Remark 3.9.5 hull/determinant source while recovering the record-canonical bridge equality through the built constructor's bridge datum, not through a new synchronization argument. The same route now has a parameterized Ob5 quotient audit. On the non-vacuous additive-Haar branch, the residue-module/unit-ball route now has a localized constructor-built C_Θ boundary audit: it records the projected Haar-modulus source, the constructor-built Remark 3.9.5 projection, the localized Step (xi) local-term summation endpoint, the local-to-global C_Θ audit, and the resulting remaining additive-Haar payload audit. For any nonempty comparison possible image, the Ob5 audit projects the constructor-built bounded-family hull inclusions, collapsed upper-semi quotient equality, determinant bound, and local canonical-scale chain. The additive-Haar source now constructs the bundled constructor-built/localized determinant-scale synchronization from arithmetic-degree calibration; the Record-Ob3/Ob5 wrapper derives the same synchronization via the record Ob3/Ob4 determinant handoff and now audits the localized Step (xi) C_Θ endpoint and residual arithmetic handoff at that boundary. Its compact-open synchronized route audit also exposes the derived signed C_Θ inequality and the Corollary 3.12 dichotomy before comparing definitionally with the established synchronized concrete endpoint. The normalized additive-Haar public route can now consume this Record-Ob3/Ob5 source directly and rebuild the localized constructor-built C_Θ route internally. The compact-open additive-Haar concrete branch now has the parallel synchronized wrapper: the packet first constructs the canonical HDD record, the Record-Ob3/Ob5 source is indexed over that canonical record, and the localized determinant-scale synchronization is projected before the synchronized concrete endpoint is invoked. The compact-open packet-to-HDD construction is now a named Lean boundary: from the compact-open log-Kummer map and pointwise calibration source it constructs the principal HDD source and canonical constructor-built source, and the synchronized concrete route now consumes this named boundary internally instead of rebuilding it inline. The lower valuation-ball Record-Ob3/Ob5 source is now packaged over this named HDD boundary: one source object carries the constructor-built/record Ob3/Ob4 determinant equality and projects the component Record-Ob3/Ob5 source, the localized determinant-scale synchronization, the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor source, the finite Haar defect, the total Haar-defect bound, and both synchronized-source and component-source forms of the pointwise Step (xi)/Haar/main equality. Thus these local inputs are no longer separate mathematical obligations at this boundary; Lean transports the pointwise equality across the component/valuation-ball formula-source identification, so the component Record-Ob3/Ob5 route sees the same decomposition. These projections are also bundled as a synchronized route-input record, giving the later concrete endpoint a single lower source artifact instead of separate component, synchronization, local-analytic, Haar-defect, and pointwise-equality inputs. The bundle also exposes the source-data determinant equality and, when the source data is the constructor-built HDD source, the corresponding component synchronization endpoint. The Theorem 3.11 possible-image nonemptiness needed for the component projection is now derived from the compact-open log-Kummer theta-region source: the principal transported theta-region cover supplies a point in every theta class, and the record/fiber-theta-region synchronization rewrites these witnesses to the record Θ-possible-image family. In the constructor-built source-data case, the same boundary is now constructed without a separate constructor-built/record Ob3/Ob4 determinant equality: that equality is read from the Record-Ob3/Ob5 valuation-ball source whose source-data field is the constructor-built HDD source. The synchronized route-input bundle is also constructed directly from the compact-open log-Kummer source and this valuation-ball source, so the named-HDD boundary no longer has to be exposed as a separate input at that layer. For the canonical-HDD compact-open endpoint, Lean now also transfers theta-possible-image nonemptiness from the primitive Theorem 3.11 record to the canonical HDD record by the direct canonical-HDD audit, constructs the canonical-HDD named boundary from the compact-open log-Kummer source plus the Record-Ob3/Ob5 valuation-ball source, and immediately threads that constructed boundary into the compact-open inverse-base-prime public route. Thus the canonical-HDD possible-image witness and named boundary are no longer public inputs at this compact-open valuation-ball handoff. This boundary has now been lowered one more source layer: Lean can construct the Record-Ob3/Ob5 valuation-ball source from the Record-Ob3/Ob5 arithmetic-divisor component source together with a Theorem 1.10 valuation-ball formula-matching source and the equality identifying its additive projection with the component formula source. The same canonical-HDD valuation-ball source now also projects the synchronized route-input bundle directly: the component Record-Ob3/Ob5 source, localized determinant/scale synchronization, Theorem 1.10 valuation-ball local-analytic source, Haar defect, and pointwise Step (xi)/Haar/main equality are read from the constructed canonical boundary before the inverse-base-prime comparison is invoked. The canonical-HDD route audit now also reconstructs, from this route-input bundle, the inverse-base-prime summand source and the pointwise local-analytic IUT IV Step (xi) C_Θ source, recording their endpoints together with the localized C_Θ handoff, signed comparison, and dichotomy at the compact-open handoff itself. A further canonical-HDD compact-open route now lowers the p-adic/additive-Haar input side to the product-level restriction-calibrated Ob7 handoff, so the structure-sheaf/direct-summand synchronization and nonzero norm-square projection are fields of the local-arithmetic source rather than separate public arguments. This route still constructs the same local-analytic C_Θ handoff from the Record-Ob3/Ob5 valuation-ball source before invoking the restriction-calibrated Step (xi) endpoint. The strongest canonical-HDD compact-open wrappers now combine both reductions: their public boundaries take the component Record-Ob3/Ob5 source, the valuation-ball formula-matching refinement, their projection equality, and either the inverse-base-prime valuation-cover/additive-Haar inputs or the product-level Ob7 handoff, then construct the Record-Ob3/Ob5 valuation-ball source internally before invoking the same signed Step (xi) endpoint. The product-handoff route has now been lowered once more: it may take the arithmetic-degree-controlled valuation-ball local arithmetic source instead of the assembled valuation-ball formula-matching record, construct that formula-matching record internally, and then apply the same component-to-valuation-ball bridge. The route-input bundle now also projects directly to the pointwise local-analytic Theorem 1.10/IUT IV localized Step (xi) source consumed by the lower C_Θ route: the bundled localized synchronization, valuation-ball local-analytic source, Haar defect, synchronized pointwise equality, and total-defect bound construct that source before the larger compact-open public endpoint is invoked. The route-input audit now records this constructed source endpoint, its localized C_Θ handoff audit, the signed C_Θ comparison, and the Corollary 3.12 dichotomy at this layer. The compact-open route-input audit also records the internally constructed inverse-base-prime summand source, the same local-analytic C_Θ source and signed endpoint, and the definitional equality between the compact-open route and the corresponding pointwise route. That source now also projects the localized C_Θ constructor source, its local-term IUT IV handoff audit, the signed bound $\Theta_{\mathrm{signed}} \le C_\Theta(-q_{\mathrm{signed}})$, and the resulting Corollary 3.12 dichotomy. These facts are therefore visible at the Theorem 1.10 valuation-ball local-analytic boundary, not only after entering a larger inverse-base-prime wrapper. The current product-hull goal-evidence audit now has the same public-surface lowering: it consumes the Theorem 1.10 valuation-ball local-analytic source and projects the localized C_Θ source, handoff audit, signed bound, and dichotomy internally. The pointwise Hodge-formula companion lowers this one step further at the same current product-hull surface: the public input is the pointwise Hodge formula source, and Lean projects the measure/summand calibration source internally before entering the Theorem 1.10 localized C_Θ handoff. A local-arithmetic companion now also constructs the restriction-calibrated source at this endpoint from the residue-module/unit-ball Haar-character valuation-cover source, the local-global realified Frobenioid restriction package, the structure-sheaf normalized Haar identities, the distinguished non-structure summand, and the theta-degree/localized-adjusted sum comparison. At the synchronized Record-Ob3/Ob5 route-input boundary, the formula-realized product-handoff route now combines this product-level Ob7 handoff with the bundled valuation-ball formula-gap data to construct the pointwise Ob7 source and the pointwise Theorem 1.10 localized Step (xi) source internally. Thus the same formula-realized route is now available at the compact-open named-HDD and component/formula boundaries: the compact-open log-Kummer map supplies the pointwise valuation-ball source, and the component/formula bridge constructs the Record-Ob3/Ob5 valuation-ball source before entering the product-handoff evidence audit. The same compact-open branch now also accepts the arithmetic-degree-controlled local source directly and derives the formula-matching source before invoking the formula-realized handoff. The component-level public route now uses this controlled/product handoff surface as well, so it no longer needs to expose the inverse-base-prime valuation cover, additive-Haar direct-summand source, local-analytic Theorem 1.10 source, Haar defect, or local Step (xi)/main-log equality. Consequently the current product-hull/Hodge-formula public theorem no longer takes the completed restriction-calibrated source itself, although its local arithmetic ingredients remain source-paper obligations. A further current-product-hull companion now constructs the assembled Theorem 1.10/IUT IV localized source from the localized Step (xi) determinant source, determinant and canonical C_Θ-scale synchronizations, valuation-ball additive-Haar local analytic arithmetic-divisor source, local Haar defects, the local Step (xi)/Haar/main-log identity, and the finite defect lower bound. Its synchronized companion lowers this further by taking a constructor-built localized Step (xi) determinant/scale synchronization source and projecting the localized determinant source and the two determinant/C_Θ-scale equalities internally. A formula-gap matched companion lowers the IUT IV input at the same boundary: the valuation-ball local analytic source is projected from named distinguished Proposition 1.4 log-shell and archimedean Proposition 1.5 metric comparison data before the localized C_Θ handoff. The source-level projection theorem now records that this Proposition 1.4/1.5-backed formula source satisfies the older Theorem 1.10 local-case endpoint once the distinguished and archimedean formula-to-gap comparisons are supplied. A route-input companion now packages the determinant/scale synchronization, formula-gap source, Haar defects, local Step (xi)/Haar/main-log identity, and defect lower bound into the typed pointwise Theorem 1.10 localized Step (xi) source, so these data are no longer separate arguments at this current-product boundary. The synchronized route-input bundle now preserves the valuation-ball formula-gap Theorem 1.10 source itself together with its local-analytic projection equality, and the pointwise inverse-base-prime public route now consumes this route-input bundle directly, so the local-analytic localized Step (xi) source is no longer a separate public input or endpoint at that boundary: the route constructs it from the bundle and then builds the formula-gap IUT IV localized source from the preserved formula-gap field consumed by the signed C_Θ comparison. The compact-open log-Kummer/valuation-cover route now makes the same handoff after deriving the principal pointwise valuation-ball source and inverse-base-prime summand source internally. Its named-HDD boundary variant goes one step lower, constructing the synchronized route-input bundle from the valuation-ball boundary before invoking this compact-open handoff. The same public route now exposes the Ob6 exact-$\Xi(P_B)$ boundary. It constructs the Remark 3.9.5 possible-image family from the constructor-built hull source, identifies the canonical $\Phi(P_B)$ approximant with the family hull, and derives the selected and comparison log-volume bounds from a source-calibrated exact-$\Xi(P_B)$ log-volume equality rather than from an arbitrary supplied approximant family. This target-charted route boundary now also has a constructor-built Ob7 product-formula source. The input is finite weighted determinant prime-strip/product-formula data plus the coric unit-character pullback; Lean requires its determinant source to be the same determinant source used by the constructor-built Step (xi) bridge, constructs the coric log-Kummer Ob7 source, and then packages the combined Ob5/Ob6/Ob7 audit. Thus the route no longer needs a standalone determinant/global Ob7 equality at this level, although the lower local product-formula and coric prime-strip data remain explicit source inputs. The determinant-calibrated public audit now also exposes the product-formula endpoint itself: the finite summand formula, determinant/global prime-strip equality, theta-derived tensor-power bound, and retained source/target/log-Kummer-column identities are named fields. At the principal-product localized public route, the Step (xi)/Remark 3.9.5 paper trace is now also constructed internally. Its Step (xi-a)–(xi-g) and Ob1–Ob9 entries are concrete statements about the unified Theorem 3.11 source spine, the $\lambda \cdot O$ hull source, the localized determinant projection, the theta-equals-family-hull equality, and the shared Ob7 log-Kummer source. The strict public route therefore no longer accepts a standalone paper-trace proposition bundle. A localized-cover variant now lowers the determinant alignment boundary: its source fixes the hull system and possible-image family to the principal-product Step (xi) source, so Lean projects the localized decomposition with the old hull-operator and possible-region equalities by reflexivity rather than accepting them as separate public inputs. Its canonical-theta refinement also takes Θ_{signed} to be the projected family-hull log-volume, making the theta/family-hull equality reflexive at this boundary. Both localized-cover forms now also have additive-payload variants: Lean constructs the paper trace, the derived Step (xi) source, and the remaining non-Step audit from the lower additive-Haar arithmetic-degree payload. The same lowering now also applies to the underlying preferred Step (xi) route ladder, from the raw Step (xi) source through the concrete-localized, principal-product, localized-cover, and transported scalar-parameter layers. The representable, locally representable, coordinate scalar-image, and real-line coordinate wrappers inherit the same additive-payload reconstruction, so these routes no longer need a separate non-Step audit when the additive-Haar payload is available. At the Additive-Haar local C_Θ boundary, the constructed Remark 3.9.5 hull/determinant source can now be built directly from the localized vector-bundle determinant source; hence the determinant-source handoff to the Step (xi) local-term constructor is reflexive. The remaining equality at this local boundary is the genuine calibration of the canonical C_Θ scale with the finite Step (xi) sum. The same constructed-trace handoff has now been propagated to the finite-extension valuation-unit-ball compact-open norm-square boundary, using the localized determinant, Hodge-evaluation, theta/family-hull, and Ob7 construction chain already required by that strict route. A calibrated variant of this boundary now replaces the raw Hodge theta-monoid/localized-adjusted sum equality by the Hodge/family-hull equality and derives the adjusted-sum equality from the localized Ob3/Ob5 identity. A further restriction-calibrated variant replaces the placewise Hodge canonical-degree/local-prime equalities by an anchored local-global restriction source together with one global Hodge canonical-degree calibration. The strongest finite-extension variant now also moves the remaining localized possible-region comparison down to the p-adic finite source itself: the equality after the unit-ball/additive-Haar/compact-open/ norm-square projections is derived by the source-core projection theorem. The principal-product finite-extension source lowers this once more: its localized possible-region field is fixed definitionally to the Theorem 3.11 principal-product possible-image family, so the strongest public wrapper no longer takes a p-adic possible-region equality as an independent input. A calibrated p-adic finite hull-cover source now also derives the pointwise Ob3 localized-region log-volume identities from local vector-bundle calibrations, instead of accepting those identities as fields of the finite-extension localized decomposition source. Its family-hull-indexed refinement derives the cover identity, pairwise disjointness, and finite-additive volume sum from an index map on the principal-product family hull. A cover-local measure-model refinement lowers the p-adic volume input further: the localized cover sum is derived from the log-volume measure model attached to the calibrated p-adic regions, and the preferred public finite-extension route can consume this measure-backed cover source directly. A direct-product refinement removes the supplied p-adic family-hull index: Lean constructs it from the local-factor cell cover of the principal-product family hull, using local-factor separation to prove the fiber and disjointness properties. A tensor-measure refinement removes the supplied p-adic cover-local measure model: direct-product cell log-volume identities and their finite tensor-cell summation formula construct the measure model used by the preferred public route. The p-adic tensor data has also been lowered to the source-core localized decomposition: Lean derives the tensor-cell log-volumes and tensor-sum formula from the adjusted determinant log-volume identities carried by that finite-extension hull/vector-bundle source. The direct-product local factors in this p-adic branch are now charted by local-ring/vector-bundle factor calibrations, so Lean projects their local-ring equality and direct-summand log-volume identities instead of taking raw factor regions at the public boundary. These chart calibrations have in turn been lowered to valuation-ball factor calibrations: the p-adic local-factor regions are now supplied by valuation-ball/additive-Haar data and then projected to the charted local-ring interface. A further principal-product compact-open-factor variant replaces the direct Hodge/family-hull equality by a localized adjusted-determinant sum calibration; Lean recovers the family-hull equality from the localized Remark 3.9.5 decomposition theorem. The additive-Haar principal-product branch now also has a cover-derived localized hull source: instead of taking the localized-region containment in the family hull and the family-hull log-volume sum as standalone fields, it assumes the explicit cover identity $\phi(\bigcup_i \Theta_i) = \bigcup_\beta H_\beta$ and the finite additivity formula for this cover; Lean derives the old containment and volume-sum interface from these lower hull-cover statements. This boundary has now been lowered again: each H_β carries a local vector-bundle calibration, is presented as a fiber of an index map on the family hull, and Lean derives the cover identity, pairwise disjointness, finite-additive cover sum, and pointwise $\log \mathrm{vol}(H_\beta)$ determinant identity after identifying that local vector bundle with the additive-Haar determinant localization. The strict positive determinant contribution has now been separated from the valuation-unit-ball normalization. A principal-product additive-Haar route constructs the paper trace internally and derives the positive summand from the source-level inequality $1 < \mu(U)$. A stronger local-field branch now constructs such a strict factor explicitly: for a finite extension of $\mathbb{Q}_p$, the inverse base-prime dilation of the normalized unit ball is identified as $p^{-1}O_K$, and the source-core Haar-modulus theorem proves $\mu(p^{-1}O_K) = p^{[K:\mathbb{Q}_p]} > 1$. Conversely, Lean proves that the older unit-ball-positive boundary is contradictory: the same unit-ball source forces $\mu(O_v) = 1$, hence normalized Haar log-volume 0. The principal-product hull source has also been connected to the lower upper-semi quotient formalism of Remark 3.9.5(v): the concrete hull source now projects to a bounded-family hull quotient source, proves that this quotient hull is the canonical/principal family hull, and proves that any two nonempty quotient-indexed possible regions have the same collapsed quotient image. For equality-orbit choices, Lean packages this with region equality, log-volume equality, quotient-image equality, and containment in the family hull. The principal-product source boundary has been lowered again in source-core: a scalar-parameter direct-product hull source now constructs the full Remark 3.9.5 principal product-hull system and its minimal intersection theorem. Product-parameter representability is no longer a global assumption: Lean derives it from coordinate scalar-image data. Once local coordinate points can be extended to product points, the coordinate landing/preimage laws prove that each coordinate product region is the image of the local integer factor under the coordinate scalar map. Lean then uses the coordinate region itself as the local scalar parameter and assembles these choices into the product scalar parameter. The intersection scalar selector is obtained from the represented direct-product intersection parameter. This replaces the principal intersection law by the direct-product hull theorem plus explicit coordinate scalar-image data. This scalar construction has now been transported onto the public Step (xi) carrier: a generic equivalence transport theorem carries principal $\lambda \cdot O$ hull systems across carrier equivalences, and the transported scalar-parameter source constructs the Point(`target`)-level principal product hull source consumed by the localized theta-equals-family-hull route. Thus the generic route no longer needs a standalone principal product-hull source on the public carrier; it needs coordinate scalar-image data and an explicit carrier equivalence. In the one-coordinate real-line specialization, even this carrier equivalence is not supplied: Point(`target`) is canonically equivalent to its real coordinate and to the product carrier Unit $\to \mathbb{R}$. The strict public coordinate and real-line endpoints now also construct the Step (xi) paper trace internally from the same coordinate scalar-image hull, localized determinant, theta-family-hull, and Ob7 data, so these endpoints no longer expose either a bare public principal-hull source or a standalone paper-trace argument. Their additive-payload variants now go one step further: Lean rebuilds the derived Step (xi) source and the remaining non-Step audit from the additive-Haar arithmetic-degree payload, matching the stricter principal-product constructed-trace boundary. The valuation-ball selected exact-Ξ boundary has also been tied to this coordinate source: Lean converts a coordinate scalar-image valuation-ball cover to the selected scalar-parameter cover by taking the local scalar parameter to be the coordinate region itself, proves that the induced holomorphic hull system agrees with the direct-product hull system, and then reuses the selected radius-cover exact-Ξ transport theorem. The coordinate source has also been lowered to a sharper local image boundary: an exact formula $H_v(\lambda_v) = \lambda_v O_v$ constructs the older coordinate landing and preimage laws, and the valuation-ball public audit now has an exact-image variant that projects through the coordinate valuation source before transporting exact-Ξ to the public carrier. Below this boundary, Lean now constructs a source-facing scalar-parameter hull system from the Hodge-theater/log-theta lattice key itself: the parameter is the tuple consisting of the Hodge theater, history label, theta-pilot lattice coordinate, and coric datum, and the local region $H_v(\lambda_v)$ is the coordinate projection of the formula-backed theta region along the public carrier equivalence. This boundary has now been lowered again: a realized local image source states that each formula-backed theta region is the realized image of its local integer factor, and that projection of the realized point is the corresponding local scalar multiple. Lean derives the projected scalar-image equality from these two local image laws and then feeds the existing scalar-parameter direct-product hull source. Independently, the exact-image direct-product formula can now be specialized to each projected theta region, constructing the same package-level coordinate scalar-image source without a standalone projected equality. For the selected valuation-ball family-hull route, the implementation now uses the more faithful scalar boundary: a nonzero local multiplication source $\lambda_v \cdot O_v$ constructs the selected scalar-parameter valuation source, and the unit-ball valuation wrapper inherits this selected source. The resulting endpoint identifies the selected scalar image with the valuation-ball direct-product cell union and with the adjusted Ob3/Ob5 determinant handoff. The same selected-scalar source now constructs the possible-image exact-$\Xi(P_B)$ source from progressively local cover data: indexed cell containment, single local-factor containment, and finally compact-open valuation-ball-radius containment, where the last step uses the additive-Haar normalization audit $B_{\beta,v} = B_{\beta,v}(r_{\beta,v})$. The nonzero-scalar and valuation-unit-ball wrappers inherit this radius-to-exact-Ξ route. Thus the exact-Ξ calibration is no longer an independent field at this boundary. The public audit packages the selected hull/cell-union equality, calibrated log-volume, normalized determinant handoff, and transported exact-Ξ source. This is still selected-parameter evidence; the remaining public-route obligation is to construct those valuation-ball containments from transported theta/valuation local exactness and to derive the realized theta-region local image laws from still lower analytic Frobenioid/log-Kummer data. For the stricter localized route, the final normalized localized determinant bound by Θ_{signed} is no longer supplied as a standalone inequality: Lean derives it from the identification of Θ_{signed} with the constructed localized family-hull log-volume. The record-canonical Remark 3.9.5 hull/determinant bridge now also feeds this record: once the paper-derived selected q-region log-volume is identified with the record bridge q-pilot log-volume, the q-region-to-normalized- determinant bound and the determinant-to-Θ_{signed} bound are both projected from the bridge. The bridge-backed paper-source constructor now derives this log-volume identification from lower alignment data: equality of the paper hull operator with the bridge hull operator, and equality of the bridge q-pilot region with the source-spine selected Theorem 3.11 possible image. The constructed-source version projects the bridge and hull operator from the constructed Remark 3.9.5 source itself, leaving only the q-pilot region/source-spine selected possible-image identification at this boundary. The Ob7 source link has now been lowered to the same record bridge: the record-canonical Remark 3.9.5 bridge supplies the q-region and determinant source, while the coric $\Theta^{\times\mu}$ invariant supplies the prime-strip/log-Kummer equality. In the same-index constructor-built possible-image route, equality of record images with source images derives both the selected q-region identification and the quotient-family union alignment. A further source-selected constructor forms the constructed Remark 3.9.5 source with the source-spine selected possible image as its q-pilot region, so this identification is definitional on that route; the remaining explicit condition is containment of the selected possible image in the record possible-image union. This selected-source boundary now also has a stricter Ob3/Ob5 determinant form: instead of receiving a raw hull/determinant compatibility record, it can receive the record-canonical family-hull compatibility source, or the still lower adjusted determinant log-volume source, and Lean derives the old compatibility projection internally. The pointwise source-region route and the current deepest retained (Ind2) action-packet transport route both have adjusted log-volume variants, so the public local source spine can pass through the measured adjusted-source boundary. That measured boundary now has a Hodge-calibrated constructor: the package measure equality is projected from the measured source, and the Θ_{signed}-to-family-hull equality is derived from 'IUTStage1HodgeFamilyHullLogVolumeCalibration', instead of being supplied as a raw downstream argument. This is still a boundary above the localized Ob3/Ob5 arithmetic-vector-bundle data used by the older compatibility routes, but the theta equality at that boundary is now Hodge-derived rather than asserted. The quotient-hull side of these variants has also been lowered: the route may now take the Theorem 3.11 possible-image quotient compatibility for the source images, and Lean forms the Remark 3.9.5 equality-quotient hull/log-volume compatibility by applying the selected holomorphic hull operator. Thus the deepest local selected route no longer needs to expose a standalone equality-quotient hull compatibility record when the lower possible-image quotient source is present. This has now been pushed down to the unified Theorem 3.11 source spine: the route can derive the possible-image quotient compatibility from the theta-pilot possible-image source together with the vertical log-Kummer packet-aligned finite-procession source. The concrete primitive packet now exposes this one-sided source directly from its own possible-image source, vertical log-Kummer source, gluing torsor, and selected q-choice. A comparison audit now relates this vertical packet source to the stricter fiber (Ind_2) action-packet transport source used by the theta-region route, proving agreement of the record possible-image family and the selected q-region as a target set. A further symmetry-label wrapper projects the fiber action-packet source from lower label-transport data tied to the packet's vertical log-Kummer indeterminacy source. The wrapper can now also be constructed from the common Theorem 3.11 possible-image witness, the equality-quotient possible-image family, and the explicit symmetry-label, upper-semi, and procession transports. The packet-facing public completion route has a corresponding possible-image-witness entry point, so it no longer exposes a preassembled symmetry-label wrapper or an arbitrary fiber (Ind_2) action-packet transport source. On this route the explicit quotient-compatibility argument disappears; the remaining source-side comparison is the record/source possible-image trace supplied by the action-packet transport layer. The determinant upper-bound input has also been lowered on the same source-spine routes: the Ob4 tensor-power bound is derived from the exact-theta calibration identifying Θ_{signed} with the record Ob3/Ob5 family-hull log-volume. Thus the strict action-packet route now uses adjusted determinant log-volume data plus exact family-hull theta calibration, rather than a standalone tensor-power estimate. The Ob7 prime-strip/log-Kummer record can now be read from a single coric $\Theta^{\times\mu}$-style source link aligned with the same quotient-hull bridge; its retained statement includes the local Gaussian/environment equalities and unit-character equalities from the Ob7 source endpoint. The strongest selected route now constructs this source link from the constructed Remark 3.9.5 bridge and a coric prime-strip invariant, after transporting determinant compatibility across the record-union/equality-quotient-family alignment. In the same-index source-spine case, this alignment is now derived from the equality of the record possible-image family with the Theorem 3.11 source image family; the selected route also derives the selected-q containment in the record union from the same equality. A lower selected route now takes the paper-shaped pointwise trace that the record possible image attached to each concrete Hodge-theater/log-theta choice equals the corresponding source-spine possible image; Lean assembles this trace into the full family equality before applying the quotient-union theorem. The selected Ob3/Ob4 route, the adjusted Ob3/Ob5 source-region route, and the exact-theta source-spine route now also have product-formula Ob7 variants: the determinant/global prime-strip equality is proved from finite local adjusted-summand calibrations and the finite prime-strip product formula, while the coric unit-character data constructs the $\Theta_{\mathrm{LGP}}^{\times\mu}$ invariant before entering the existing log-Kummer Ob7 source link. These routes therefore no longer need the determinant/global Ob7 equality as an independent input. This pointwise trace is now itself derived from the existing theta-pilot class-region source once its class formula is identified with the Hodge-theater source spine's theta-pilot class-image family. The route now also accepts record-level theta-pilot class formulas and Hodge-theater/history/log-theta-lattice formulas, projecting them through the class-region source before assembling the Step (xi) comparison. Below the lattice formula layer, it can consume the same-lattice-key image law, which constructs the lattice formula from representative choices before comparison with the source-spine class images. One level lower, it can consume typed theta-pilot fiber-transport invariance and derive the same-key image law internally. Below that, the selected route can now consume the (Ind1), (Ind2) equality-quotient pullback and, on the current strongest local source route, the retained post-procession (Ind2) action-packet transport source; Lean derives the quotient image pullback, the fiber-invariant lattice image law, and the record/source trace before entering the Step (xi) hull/determinant comparison. The adjusted Ob3/Ob5 determinant-log-volume variants have now been added at the class-region, class-formula, lattice-formula, lattice-image-law, and typed fiber-transport levels, so these intermediate routes inherit the record-canonical determinant compatibility from the adjusted log-volume source rather than exposing a raw hull-approximant compatibility record. The action-packet theta-formula comparison with the source-spine class images can now also be derived from the equality of the record possible-image family with the unified Theorem 3.11 source image family; it is no longer a separate input on the exact-theta action-packet route. A further constructed-record variant installs the unified source-spine image family directly as the record's theta-pilot possible-image family, so the record/source image equality becomes definitional; the remaining image-side calibration is that these source-spine images are the package target-region family. This calibration has now been lowered from a whole-family equality to the pointwise theta-pilot class-region trace $\mathcal{R}_{\mathrm{class}}(\mathrm{class}(c)) = \mathcal{R}_{\mathrm{target}}(c)$; Lean derives the source-image/target-region family equality internally before installing the source-spine record. One layer lower, the class-region trace can now be produced from a target-region-backed theta-pilot source: a representative section for theta-pilot classes, a proof that target regions depend only on the theta-pilot class, and explicit target-region point witnesses. Lean constructs the witness-bearing class-indexed possible-image source from these data and threads it through the exact-theta Step (xi) action-packet route. This has now been lowered once more: the target-region class-invariance law can be derived from invariance under the concrete theta-pilot lattice key, namely Hodge theater, history label, log-theta lattice coordinate, and coric datum. This lattice-key boundary has itself now been lowered to a target-region lattice formula: each package target region is identified with a formula indexed by the Hodge theater, history label, theta-pilot lattice coordinate, and coric datum, and Lean derives the key invariance, class trace, full source-image/target-region family equality, and Step (xi) source-spine handoff from that formula. The formula is now also produced from the concrete lattice-image law already present in the Theorem 3.11 source spine: record possible images are evaluated by the lattice formula, and the multiradial record's target-region realization identifies those record regions with the package target regions. The exact-theta Step (xi) source-spine route can therefore consume the lattice-image law directly, without an independently supplied target-region lattice formula. The same package target-region formula is now projected further down from typed theta-pilot fiber transport, from the (Ind1), (Ind2) equality-quotient pullback, and from the retained post-procession (Ind2) action-packet transport source. Thus this target-region formula boundary follows the current deepest local Theorem 3.11 action-packet source rather than stopping at a standalone lattice-image law. The action-packet boundary itself has now been lowered one step: Lean projects the retained packet/action transport source from place-audited nonarchimedean Ism or archimedean order-two action-entry label-transport data, together with explicit identifications of the audited packet tensor states with the post-procession source and target local tensor states. The label-transport source is now constructed one layer lower from the retained action-entry source, source and target tensor symmetry label equations, and transport of the direct-summand symmetry kind. The same tensor-label data now also projects to the place-audited possible-image/fiber audits for the nonarchimedean and archimedean branches, so the lower action data carries both the packet transport and the equality-quotient/fiber witness information. This boundary has now been lowered again: the tensor-label equalities may be read from a single direct-summand packet SymmetryLabelTransportSource, and the corresponding concrete fiber source projects through the tensor-label route internally. Record-level possible-image point witnesses now also factor through the class-indexed Theorem 3.11 possible-image source and the record/source image equality, so the newest symmetry-label fiber source no longer needs independently supplied witness points. The same record/source image equality now gives all-choice nonemptiness for the additive-Haar arithmetic-divisor and record Ob3/Ob5 constructors, and builds the central Theorem 3.11 possible-image source-data record with its nonemptiness audit, rather than requiring a separate downstream nonemptiness field. The exact-theta source-spine/action-packet Step (xi) route has also been lowered at the hull/determinant obligation boundary: the package bridge equality, q-pilot positivity, and normalization are now projected from a single ‘IUTStage1SourceHullDetObligations‘ object, with an endpoint recording the induced record-canonical selected source-spine bridge. The same exact-theta source-spine route now has a localized hull/vector-bundle entry point: after an explicit identification of the localized possible-image family with the source-spine ‘recordThetaPossibleImage‘ family, Lean constructs the record Ob3/Ob5 adjusted determinant source internally and records the induced finite adjusted-sum, normalized determinant, and bridge-synchronization facts at the endpoint. This localized handoff has also been lifted through the tensor-label, label-transport, and symmetry-label transport sources, each of which projects the Fiber-Ind2 action packet internally, and through the concrete target-region lattice-formula source, which installs the package target-region possible images before entering the localized determinant constructor. It now also consumes the concrete lattice-image law itself, deriving the target-region lattice formula before entering the same localized hull/vector-bundle path. At the principal valuation-ball boundary, the symmetry-label packet now has a source object that takes a source-core Remark 3.9.5 principal valuation-ball cover together with the fiber-transport-backed family-union comparison and projects both the theta-region principal source and its principal hull. This removes a lower possible-region matching datum at the source layer. The packet/symmetry-label public route now consumes this source object directly: the theta-region principal source and selected q-choice are projected from the same packet source before entering the constructed Step (xi) completion audit. The possible-image-witness variant now constructs the packet-indexed symmetry-label wrapper before entering this principal valuation-ball route, so the preassembled wrapper is no longer public at this surface either. The same lowering now reaches the HDD-source audit route: the principal-product Ob3/Ob5 measure-Hodge-gap HDD source remains explicit there, but the symmetry-label wrapper is constructed from possible-image witness data before the HDD source is audited against the theta-region principal hull. The same witness-derived entry point now reaches the constructed-HDD route as well: Lean builds this HDD source internally from the theta-region source, packet selected q-choice, constructed qualitative bundle, measure calibration, side conditions, and summand-Hodge data, so neither a preassembled symmetry-label wrapper nor an independent principal HDD source remains at that route surface. The witness-derived lowering now also reaches the localized finite-place data route: the localized determinant source, main-log contribution, Haar defect, and summation identities remain explicit, but the symmetry-label wrapper is again constructed from possible-image witness data before the localized C_Θ source is built. The remaining principal-product source still visible there belongs to the restriction-calibrated local arithmetic boundary. The newest localized calibration variant combines both lowerings: after constructing the symmetry-label wrapper from possible-image witness data, it lowers the two calibration inputs to one theta-region calibration source. Its endpoint records pointwise pre-ledger volume as the principal holomorphic-hull log-volume and the Hodge theta-monoid degree as the finite adjusted determinant summand sum, and Lean projects the older measure and summand-charted calibrations from these formulas. The local arithmetic side is now lowered in parallel, including on the witness-derived surface: the restriction-calibrated Frobenioid/residue-module source, the structure-sheaf/direct-summand synchronization, and the nonzero norm-square Ob7 summand first define an audited local handoff source, and the possible-image witness route constructs the symmetry-label wrapper before consuming this handoff. The same witness-derived branch now projects the principal hull used by local arithmetic from the theta-region source, instead of accepting an independent principal-product hull argument, and its fiber-backed variant constructs the theta-region source from the principal valuation-ball product-hull cover and transported family-union comparison after deriving the symmetry-label wrapper from possible-image witness data; the theta-region-defined variant then projects those two inputs from a single principal theta-region valuation-ball source on the same witness-derived surface, and the pointwise metric variant now carries that witness-derived lowering through the constructed-log-shell/metric-zero valuation-ball source; the pointwise calibration variant projects the theta-region measure/summand calibration from the same witness-derived surface; the pointwise local arithmetic variant also projects the restriction-calibrated Ob7 handoff from that surface's pointwise source data, and the localized Step (xi) variant projects the determinant/summation/Haar-defect source from the same witness-derived pointwise boundary; the IUT IV arithmetic-divisor variant projects that localized source from the local evaluation ledger on the same surface, and the Theorem 1.10 valuation-ball variant projects that ledger from the valuation-ball additive-Haar formula-gap source on the same surface; the Hodge-formula variant also constructs the pointwise measure/summand calibration from explicit formulas on that witness-derived surface, and the local-analytic variant checks the valuation-ball local-analytic source on the same surface before projecting to the Theorem 1.10 valuation-ball route. The strict pointwise local-analytic route now constructs the packet-facing Ob7 source from this product handoff while projecting the formula-gap matched IUT IV source from the local analytic valuation-ball source. Its compact-open companion constructs the principal pointwise valuation-ball source from compact-open log-Kummer map data and uses the same product-handoff local arithmetic fields; a pointwise formula-gap companion now reconstructs the local-analytic Step (xi) wrapper from the same localized Theorem 1.10/IUT IV source used by the signed C_Θ route at this local arithmetic boundary, and the formula-realized evidence route performs the same compact-open and Ob7-handoff construction before entering the pointwise Theorem 1.10/IUT IV packet comparison. Its product-handoff version packages the local-arithmetic fields into the typed Ob7 handoff at the public boundary. The local-arithmetic-degree version also constructs the pointwise Theorem 1.10/IUT IV localized source itself from the localized local-term Step (xi) data and formula-gap arithmetic-degree identities. Its synchronized compact-open variant constructs that localized local-term source from determinant synchronization and finite-place arithmetic-degree data before entering the same signed route. The lower constructor route can rebuild that pointwise source from the restriction-calibrated residue-module/unit-ball/additive-Haar source plus the two projection facts before the localized route is entered. The compact-open constructed-synchronization variant also builds the determinant/scale synchronization source from the localized determinant source and its two constructor-built comparison identities. The explicit-formula compact-open variant then constructs the measure/summand formula record itself from the three displayed formula identities before the same route is entered. A determinant-calibrated refinement lowers the Hodge summand boundary further: it takes the charted Hodge calibration to the normalized theta-region determinant, and Lean derives the determinant log-volume equality and the finite adjusted-summand formula from the record-native determinant source. The realized-exact variant constructs the compact-open log-Kummer map source from realized compact-open theta-region image exactness before applying that route. The restriction-calibrated variant constructs the product Ob7 handoff from the lower local-arithmetic data and derives the Theorem 1.10 formula-gap valuation-ball source from local-analytic arithmetic-divisor data before entering the same path. The current compact-open map route now consumes that lower pointwise restriction-calibrated local-arithmetic source directly and constructs the Ob7 handoff internally. Its p-adic finite-extension refinement also projects the ordinary localized Step (xi) determinant from the unit-ball/valuation-Haar compact-open norm-square localized determinant layer at this public boundary. The synchronized p-adic finite refinement bundles that projected determinant with the constructor-built determinant and canonical C_Θ-scale identifications as a single source object before entering the local-upper endpoint. These synchronization facts remain lower source data, but they no longer appear as two separate public route arguments. The formula-gap variant now consumes the Theorem 1.10 valuation-ball additive-Haar formula-gap source and projects the local-analytic arithmetic-divisor ledger internally for the residual calculation. The current local-analytic residual refinement lowers this boundary again: the public endpoint supplies the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger and residual Haar bound, and Lean constructs the formula-gap residual source internally. The localized residual refinement also removes the p-adic determinant/scale synchronization record from the strongest endpoint: it now supplies the finite-extension localized Step (xi) determinant source and the two constructor-built determinant/canonical-scale comparison facts, from which Lean constructs the synchronization object. A principal-product constructor pushes this determinant boundary lower again by projecting the p-adic localized Ob3/Ob4 determinant source from the Theorem 3.11 principal-product localized hull-vector-bundle decomposition before entering the same local-analytic residual route. The corresponding public route now threads this principal-product residual source through the source-hypothesis endpoint: the residual input is tied to the packet's source-data-with-target-regions object and to the principal hull projected from the internally constructed theta-region valuation-ball source. Its realized-exact variant projects the compact-open log-Kummer map source from compact-open realized exactness before entering the same principal-product residual handoff; the localized principal-product source now also separates the finite-place index universe from the principal-hull parameter universe. A local-arithmetic-handoff variant now constructs the pointwise restriction-calibrated Ob7 source from the product-level local arithmetic handoff before entering that residual endpoint. A formula-gap residual variant now lowers the IUT IV input again: it takes the valuation-ball additive-Haar formula-gap residual source and projects the local-analytic residual ledger internally. A projected-principal-product refinement now constructs the p-adic principal-product finite formula-gap residual package from the product-level local-arithmetic handoff, the determinant/scale equalities, and the formula-gap residual data before entering that endpoint. Its local-analytic companion promotes the Theorem 1.10 valuation-ball local-analytic arithmetic-divisor ledger to the formula-gap matched IUT IV source internally before the same principal-product projection. The formula-gap principal-product source now also constructs the q-normalized projected-principal local-analytic source directly: Lean first projects the formula-gap Theorem 1.10 data to the local-analytic ledger, then applies the principal-product canonical-scale synchronization theorem to obtain the normalized theta identity and the signed dichotomy. The principal-product localized local-analytic source now also defines the finite-place Haar-normalization defect directly as the residual $U_v - \mathrm{StepXI}_v - M_v$, where U_v is the IUT IV local arithmetic-upper contribution, StepXI_v is the projected localized determinant summand, and M_v is the local main-log contribution. Lean proves the resulting $\mathrm{StepXI}_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$ equality by expanding the arithmetic-upper ledger and separately recovers the total Haar lower bound from the source residual hypothesis. Both the projected-principal local-analytic source and its formula-gap refinement now expose the constructor-built localized C_Θ source directly, including its endpoint, IUT IV handoff audit, signed bound, and dichotomy. A q-normalized p-adic localized residual source now lowers the generic scale boundary: it records $\theta_{\mathrm{signed}} = (\sum_v \mathrm{adeg}_v^{\mathrm{adj}})(-q_{\mathrm{signed}})$, and Lean derives the canonical finite-sum scale equality from q-pilot positivity. Conversely, the determinant/scale synchronization source now derives this normalized theta identity from the canonical finite-sum scale equality and q-pilot positivity; the generic localized local-analytic residual source therefore constructs the q-normalized source internally. A projected-principal specialization applies the same derivation to the principal-product p-adic localized determinant projection; the principal-product canonical-scale source now constructs the corresponding q-normalized projected-principal source internally before reading off its endpoint and dichotomy. The reduced projected-principal local-analytic endpoint now consumes this q-normalized source instead of a direct canonical-scale equality. The determinant side has also been lowered one level: an adjusted-determinant synchronized q-normalized source identifies the constructed principal HDD Ob3/Ob4 adjusted determinant source with the adjusted determinant source projected from the p-adic finite localized Step (xi) object, and Lean derives the weighted determinant equality used by the localized C_Θ handoff. The same adjusted-determinant source now also exposes the $C_\Theta \geq -1$ companion endpoint by projecting through the q-normalized local-analytic handoff. A heterogeneous projected-weighted determinant source now handles the endpoint case in which the principal HDD determinant side and the localized p-adic side have different hidden γ-indices: it compares the projected weighted determinant summands, anchor, and positive tensor power over the common β-index, then derives the weighted determinant equality before entering the q-normalized handoff. The public realized-exact local-arithmetic route now consumes this heterogeneous source and projects the q-normalized residual handoff internally, so the determinant equality is no longer a public endpoint input on that route. At the same heterogeneous local-analytic boundary, the residual Haar inequality can now be constructed from the packaged local arithmetic-degree residual source rather than supplied as a raw total bound. This local-analytic endpoint now also has a side-condition-sourced variant: q-pilot positivity and source normalization are supplied as `IUTStage1SourceSideConditions`, and Lean reconstructs the named side-hypothesis record internally before applying the heterogeneous projected-weighted route. The heterogeneous projected-weighted source now also exposes the $C_\Theta \geq -1$ companion endpoint by projecting through the same q-normalized local-analytic handoff, matching the signed endpoint at the non-same-index determinant boundary. The IUT IV side of this heterogeneous boundary has now been lowered as well: the ProjectedWeightedDeterminantQNormalizedFormulaGapResidualHaarSource variant consumes the Theorem 1.10 formula-gap arithmetic-divisor source, derives the local-analytic ledger internally, and exposes both the signed dichotomy and the $C_\Theta \geq -1$ endpoint at the same projected determinant boundary. Its constructor now also derives the formula-gap residual Haar inequality from the packaged local arithmetic-degree residual source, rather than taking that finite-sum inequality as a separate source field. The same constructor now feeds a direct endpoint theorem returning both the signed dichotomy and the $C_\Theta \geq -1$ companion bound from that local-degree payload. This lowering is now visible on the realized-exact local-arithmetic public route itself: the formula-gap projected-weighted wrapper first constructs the local-analytic projected source internally and then enters the existing heterogeneous endpoint, so the local-analytic IUT IV ledger is no longer the route-level input on that path. The same public wrapper now has the $C_\Theta \geq -1$ companion theorem, and both the signed theorem and this companion have now been lifted to the direct side-condition surface. Thus the public route can expose the signed dichotomy and lower-bound endpoint at one formula-gap projected determinant boundary without a separate side-hypothesis wrapper. A constructor-level public endpoint lowers this boundary once more: instead of taking the projected formula-gap source as a route argument, it takes the principal-product local Step (xi) decomposition, the projected weighted summand/anchor/tensor-power identifications, the q-normalized theta identity, the Theorem 1.10 formula-gap valuation-ball source, and the local arithmetic-degree residual payload. Lean derives the finite residual Haar lower bound from that payload, constructs the projected formula-gap source internally, and returns both the signed dichotomy and $C_\Theta \geq -1$. The same constructor-level endpoint now also has a side-condition-sourced variant: q-pilot positivity and source normalization are supplied as 'IUTStage1SourceSideConditions', and Lean rebuilds the named side-hypothesis record internally before entering the constituent route. Its local-degree side-condition form combines both lowerings: the direct side-condition surface no longer takes the residual Haar lower bound as primitive, but derives it from the local arithmetic-degree residual payload. The projected side-condition surface now also has a case-bounded form: the valuation-ball distal/non-distal/archimedean residual source is projected internally to the local arithmetic-degree residual payload before the signed dichotomy and $C_\Theta \geq -1$ endpoint are applied. The stricter same-index concrete source now records the case where the local-arithmetic handoff, the p-adic localized Step (xi) determinant source, and the IUT IV local-analytic ledger share the endpoint β-index. Lean now projects the q-normalized principal-product residual source from this concrete source and derives the signed dichotomy without passing through the older public wrapper. This same-index route is now also exposed as the preferred public endpoint for the shared-β case, so the public theorem surface consumes the constructed same-index source directly rather than a separate localized C_Θ source, determinant equality, or raw signed comparison. The same signed endpoint now also has a side-condition-sourced variant: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` before applying the constructed same-index source route. Its route audit records the q-normalized endpoint, the localized C_Θ handoff audit, the derived IUT IV signed upper bound, and the resulting Corollary 3.12 dichotomy from the same source object. The companion goal-evidence audit gives the corresponding relative nonvacuity statement: an inhabited same-index source inhabits the route-audit proposition and yields the same signed endpoint. This does not claim construction of the foundational Hodge-theater/Frobenioid data themselves. A canonical-scale same-index companion now removes the raw normalized theta identity from this local-analytic source boundary. Its public source is the principal-product localized residual package carrying the determinant equality, canonical C_Θ-scale equality, Theorem 1.10 local-analytic ledger, and residual Haar lower bound; Lean then applies the principal-product determinant/scale synchronization theorem to construct the q-normalized projected-principal source internally. Thus this route obtains $\theta_{\text{signed}} = (\sum_v \operatorname{adeg}_v^{\text{adj}})(-q_{\text{signed}})$ from canonical scale synchronization rather than postulating it as a same-index field, and the same internally q-normalized route supplies the $C_\Theta \geq -1$ companion endpoint. This canonical-scale route now also has signed and $C_\Theta \geq -1$ side-condition-sourced public endpoints: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` before entering the same scale-synchronized local-analytic source route. A formula-gap same-index companion now lifts this replacement through the Theorem 1.10 formula-gap boundary. Its source carries the principal-product formula-gap residual package; Lean projects the local-analytic ledger and then uses the same canonical-scale route to obtain the signed endpoint, again without a raw formula-gap theta-normalization field. The corresponding

$C_\Theta \geq -1$ endpoint is now read from this same internally q-normalized scale-synchronized route. The formula-gap canonical-scale route now has the same side-condition-sourced signed and $C_\Theta \geq -1$ variants, so the public surface can consume `IUTStage1SourceSideConditions` rather than a separate side-hypothesis wrapper at this boundary. The case-bounded version of this scale-synchronized route now constructs the formula-gap residual package from the scale-synchronized local-analytic source, the valuation-ball formula-gap source, and the explicit distinguished, nondistinguished, archimedean, and distinguished +1 Theorem 1.10 case bounds. Thus the valuation-ball case split feeds the formula-gap endpoint without reintroducing a raw same-index theta-normalization field. The scale-synchronized local-analytic residual source itself now has a lower constructor from local arithmetic-degree residual data. Its determinant equality and canonical C_Θ-scale equality still enter at this boundary, but the total Haar residual lower bound is no longer a separate field: it is read from the packaged local arithmetic-degree residual source, whose pointwise $StepXI_v + Haar_v$ identity has already been summed over the finite place index. The corresponding $C_\Theta \geq -1$ endpoint is also exposed directly from this internally q-normalized scale-synchronized source. This lowering has also been lifted to the concrete same-index public route. Lean now constructs the projected principal-product p-adic finite Step (xi) source from the realized-exact local-arithmetic handoff and assembles the same-index scale-synchronized local-analytic source from the remaining determinant/scale synchronization facts, the Theorem 1.10 local-analytic ledger, and the packaged local arithmetic-degree residual source. Thus the signed dichotomy and $C_\Theta \geq -1$ endpoint are available without taking the scale-synchronized local-analytic source itself as an atomic public input. The determinant side of this constructor has now been lowered further: Lean derives the constructor-built weighted determinant equality from the three component equalities for weighted summands, anchor, and positive tensor power against the projected localized Step (xi) determinant source. The canonical C_Θ-scale equality remains the explicit scale-synchronization input for this constructor path, but the canonical audited boundary is now the scale-synchronized source itself: the whole determinant-source synchronization and canonical scale synchronization are the remaining source facts, while the three component equalities are only one checked way to construct the former. At the packaged q-normalized local arithmetic-degree boundary, Lean now also derives this canonical scale equality from the theta/log-volume identity $\theta = \sum_v StepXI_v \cdot (-q)$ together with q-pilot positivity before entering the scale-synchronized local-analytic route. The same-index formula-gap refinement now lowers the IUT IV side of this source: it consumes the Theorem 1.10 formula-gap matched arithmetic-divisor source and projects the local-analytic ledger internally before entering the same q-normalized route. At this boundary the residual Haar lower bound is now also derived from local arithmetic-degree data: Lean proves that the pointwise identity $StepXI_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$, together with the total Haar-defect lower bound, gives the formula-gap residual inequality used by the same-index endpoint. The reusable local source IUTStage1IUTIVLocalArithmeticDegreeResidualSource packages this payload independently of the same-index route: it exposes the local arithmetic-upper identity, the arithmetic-upper finite sum as $\sum_v (StepXI_v + Haar_v + M_v)$, the main-log finite sum, the residual lower bound, and an endpoint audit from the same local degree and Haar-defect data. The same-index namespace now has the corresponding residual-field constructor from this packaged payload, so the formula-gap source can fill its residual inequality from the local arithmetic-degree residual source rather than from separate raw Haar-defect and pointwise equality inputs. The same payload now also constructs the constructor-built localized Step (xi) local-term C_Θ source from the determinant/scale synchronization source and the IUT IV local evaluation ledger, eliminating the separate local main-log, Haar-defect, arithmetic-upper sum, and main-log sum arguments on that constructor path. A stricter residual layer IUTStage1IUTIVLocalArithmeticDegreeDefinedResidualSource now defines the Haar defect itself as

$$\mathrm{localArithmeticUpper}_v - \mathrm{StepXI}_v - \mathrm{mainLog}_v.$$

Lean proves the pointwise identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ by unfolding the IUT IV local arithmetic-upper contribution, and then constructs the older packaged residual source and the localized Step (xi) local-term C_Θ source from this defined-residual input. Thus the Haar-defect function and pointwise equality are no longer independent data at this lowered layer; the remaining numerical input is the global lower bound for the defined residual sum. A pointwise lower source now reduces that global input to local nonnegativity of every defined residual together with one distinguished local residual contribution at least 1; Lean derives the finite summed lower bound by splitting the finite sum into the distinguished term and the remaining nonnegative terms. The same-index formula-gap residual theorem now also consumes this defined-residual source directly, projecting internally to the packaged residual source before entering the existing q-normalized route. The parallel same-index source ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedDefinedResidualSource uses the same determinant and theta-equality data as the local-degree source but exposes the defined-residual payload at its boundary, and Lean projects it to the existing local arithmetic-degree source internally. This projection is now exposed all the way to the named preferred public endpoint and its route audits: the defined-residual same-index source constructs the formula-gap source, the q-normalized local-analytic source, the signed Corollary 3.12 dichotomy, and the public-boundary inventory by internal projection to the local arithmetic-degree route. The same projection now also reaches the scale-synchronized local-analytic endpoint: the local arithmetic-degree stage derives the canonical C_Θ-scale equality from the q-normalized theta/log-volume identity and q-pilot positivity, so the defined-residual boundary does not carry that scale equality as a separate public input. Thus the packaged residual source is no longer a public input at this endpoint; the remaining local-estimate trust boundary is the lower bound for the globally summed defined residual. The next parallel source, ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedPointwiseDefinedResidua lowers this boundary at the same public endpoint: it exposes the pointwise nonnegativity/distinguished-residual source, constructs the summed defined-residual source internally, and then enters the defined-residual public route. The endpoint therefore no longer takes the global residual lower bound as a raw same-index input; what remains is to derive the pointwise local nonnegativity and distinguished contribution from the nonarchimedean, non-distal, and archimedean IUT IV local estimates. This lower local-estimate boundary has now been attacked directly. The source IUTStage1IUTIVLocalArithmeticDegreeGapBoundedPointwiseResidualSource derives the pointwise residual source from the local arithmetic-gap inequalities

$$\mathrm{StepXI}_v \leq \mathrm{localArithmeticUpper}_v - \mathrm{mainLog}_v$$

for all places, together with one distinguished $\mathrm{StepXI}_{v_0} + 1$ version of the same inequality. Lean proves the residual nonnegativity and the distinguished residual lower bound by ordered real arithmetic. The further source IUTStage1IUTIVTheorem110ProcessionBoundPointwiseResidualSource uses the case-defined Theorem 1.10 procession upper bound: once localized Step (xi) is bounded by that procession bound, the existing Theorem 1.10 formula source supplies the comparison with localArithmeticUpper $-$ mainLog, and Lean constructs the pointwise defined-residual source internally. The strongest same-index public route has since been rewired past the pointwise residual wrapper through the valuation-ball case-bounded same-index source described below; the remaining local-estimate work is to derive those distinguished, nondistinguished, and archimedean case comparisons from the still lower IUT IV estimates rather than from a supplied case-bounded residual source. The valuation-ball formula-gap layer now has the same lowering: the IUT IV source IUTStage1IUTIVTheorem110ValuationBallAdditiveHaarFormulaGapMatchedArithmeticDivisorEvaluationSource projects directly to a formula arithmetic-divisor source, and IUTStage1IUTIVTheorem110ValuationBallProcessionBoundPointwiseResidualSource uses that projection to derive the same pointwise residual source from the valuation-ball additive-Haar Theorem 1.10 data. This has now been split into the actual local cases visible in IUT IV Theorem 1.10. The source IUTStage1IUTIVTheorem110ValuationBallCaseBoundedPointwiseResidualSource replaces the uniform comparison with the projected procession bound by the distinguished nonarchimedean bound, the nondistinguished nonarchimedean zero bound, the archimedean bound, and one distinguished +1 inequality. Lean then reconstructs the projected case-defined procession inequality and projects to the valuation-ball procession residual source and the pointwise defined-residual source internally. The same case-split source now also feeds the localized Step (xi) local-term C_Θ constructor: Lean first projects the valuation-ball local cases to the pointwise residual source, then to the defined residual source, and finally constructs the localized C_Θ source without taking the pointwise residual wrapper as a separate input at that handoff. The closed formula-gap variant constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallFormulaGapCaseBoundedResidualSourc lowers the same handoff one step further: its local-analytic and Theorem 1.10 C_Θ trace records are projected from the valuation-ball additive-Haar formula-gap source, rather than being supplied as separate constructor arguments. The local-analytic companion constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticCaseBoundedResidualSou pushes this lower again: it constructs the formula-gap matched Theorem 1.10 source from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before entering the same case-bounded C_Θ handoff. The new local-analytic case-source form constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticCaseSourceResidualSourc removes the formula-gap-indexed case residual from this localized handoff as well: the distinguished, nondistinguished, archimedean, and distinguished +1 inequalities are stated directly over the local-analytic valuation-ball source, and Lean constructs both the formula-gap matched source and the older case-bounded residual source internally. The case-split residual source also now projects directly to the defined residual source and the packaged local arithmetic-degree residual source, so downstream constructors can consume the summed residual and Haar-defect package without reintroducing a pointwise residual argument. The newest local-analytic refinement lowers this boundary once more. The source IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionBoundPointwiseResidualSource does not carry the three expanded case inequalities. It carries the single comparison of localized Step (xi) with the projected case-defined Theorem 1.10 procession upper bound, plus the distinguished +1 comparison and the distinguished-kind witness. Lean expands the local case definition to recover the older local-analytic case-bounded source and then applies the same localized C_Θ constructor through constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticProcessionBoundResidual Thus the remaining Step (xi)-to-procession comparison is still a genuine source boundary, but the separate distinguished/nondistinguished/archimedean case fields are no longer independent inputs at this local-analytic handoff. That comparison has now been factored into its paper-facing components. The source IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionComparisonPointwiseResidualSource introduces a procession-normalized local log-volume, identifies the localized Step (xi) weighted adjusted log-volume with it, and separately records the procession-normalized upper-semi estimate against the projected Theorem 1.10 procession bound. Lean derives the previous Step (xi)-to-procession source and the localized C_Θ constructor constructorBuiltLocalizedStepXILocalTermCThetaSourceOfValuationBallLocalAnalyticProcessionComparisonRes from these two ingredients. The upper-semi estimate has now been lowered again by IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalEstimatePointwiseResidualSource: Lean identifies the normalized local procession value case-by-case with the distinguished Step (v) exact procession bound, the nondistinguished Step (vi) zero tensor bound, or the archimedean Step (vii) metric-container log-volume, then derives the projected Theorem 1.10 processional upper bound from the formalized Proposition 1.4/1.5 inequalities. The remaining local boundary is the Step (xi) identification with this normalized procession quantity together with the distinguished one-unit residual inequality. A companion route, IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalArithmeticGapPointwiseResidualSource, keeps the processional estimates for the ordinary local bounds but sends the distinguished one-unit term through the arithmetic-minus-main residual inequality used in the Haar-defect summation. Thus the localized C_Θ handoff no longer requires the stronger assertion that the distinguished +1 margin lies below the local procession formula. This boundary has now been lowered one step further by IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalPointwiseResidualSource: it takes the pointwise defined Haar-residual source $1 \leq$ arithmeticUpper $-$ StepXI $-$ mainLog at the distinguished place, identifies StepXI with the procession-normalized local log-volume, and derives the distinguished arithmetic-minus-main +1 comparison internally before projecting to the arithmetic-gap route. The next Lean layer, IUTStage1IUTIVLocalArithmeticDegreePointwiseHaarDefectResidualSource, identifies the defined residual with the local Haar-normalization defect via the arithmetic-degree identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$. Its processional wrapper IUTStage1IUTIVTheorem110ValuationBallLocalAnalyticProcessionalHaarDefectResidualSource therefore constructs the pointwise residual source from local Haar-defect nonnegativity and one distinguished defect lower bound, rather than taking the pointwise residual inequalities themselves as primitive. A p-adic unit-ball refinement now lowers this Haar-defect boundary: IUTStage1FinitePlacePadicUnitBallHaarIndexDefectSource defines the local defect as $p_v^{[K_v:\mathbb{Q}_{p_v}]}$ from the unit-ball Haar-character normalization source, proves local nonnegativity and the distinguished one-unit lower bound, and records the source Haar endpoint $\mu(O_v) = 1$, $\mu(p_v O_v) = p_v^{-[K_v:\mathbb{Q}_{p_v}]}$. The new IUTStage1IUTIVLocalArithmeticDegreePadicUnitBallHaarDefectResidualSource and its processional local-analytic wrapper project to the older Haar-defect route before entering the localized C_Θ constructor. This boundary is now lowered once more by the Step (xi) synchronization source: it assumes the local calibration

$$StepXI_v = A_v - M_v - p_v^{[K_v:\mathbb{Q}_{p_v}]},$$

where A_v is the already-constructed local IUT IV arithmetic upper term. Lean unfolds $A_v = a_v(D_v + C_v) + E_v$ to recover the expanded arithmetic-degree identity internally. This synchronization is now further lowered to the formula matching equalities $\mathrm{StepXI}_v = a_v(D_v + C_v)$ and $E_v = p_v^{[K_v:\mathbb{Q}_{p_v}]} + M_v$. The stricter audited bridge from the formula-gap matched valuation-ball arithmetic-degree/p-adic source now projects those source-backed formula pieces to the canonical Step (xi) constructor: the Step (xi) equality comes from localized determinant weight and raw-log-volume calibration, the prime-error equality is routed through the p-adic Haar-index defect source, and the formula-gap provenance is preserved. Thus the remaining trust boundary is the derivation of those calibration and prime-error inputs from the full IUT IV local estimate construction, not the Haar-defect lower bound or an independent synchronization record. The same case split has now been lifted to the same-index signed boundary by ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedCaseBoundedResidualSou It keeps the determinant and theta-equality fields of the pointwise same-index route, but its residual payload is the valuation-ball Theorem 1.10 case split itself. Lean projects this source to the pointwise residual source, then to the q-normalized local-analytic source, where it constructs the signed C_Θ bound object. The named case-bounded public endpoints now expose both the signed Corollary 3.12 dichotomy and the companion $(-1 : \mathbb{R}) \le C_\Theta$ lower bound, so the pointwise residual package is no longer the last public same-index residual input on that route. The same route is now pinned by endpoint-level evidence and public-boundary inventory audits: Lean records that the case-bounded valuation-ball source first constructs the pointwise residual source, then the defined residual and q-normalized comparison data, before reading off the dichotomy. The boundary has now been lowered once more by the source ConcretePacketProjectedPrincipalProductSameIndexWeightedDeterminantQNormalizedLocalArithmeticDegreeSo it has the same same-index determinant and theta-equality fields as the formula-gap source, but replaces the raw residual inequality field by the packaged local arithmetic-degree residual source. Lean constructs the older formula-gap source and hence the local-analytic q-normalized route from this local-degree source, and the same namespace exposes the resulting signed Corollary 3.12 dichotomy directly from that local-degree source. Its route audit now also exposes, from the same local-degree source, the constructed q-normalized endpoint, localized C_Θ audit, signed IUT IV upper bound, and dichotomy. The named preferred public endpoint now has the corresponding local-degree version, so the signed endpoint can be invoked without presenting the older local-analytic source as the public input. There is also a side-condition-sourced local-degree endpoint: Lean rebuilds the named side-hypothesis record from `IUTStage1SourceSideConditions` and returns both the signed dichotomy and the companion $C_\Theta \ge -1$ bound from the same local-degree source, using the formula-gap projection for the dichotomy and the scale-synchronized projection for the lower bound. The next wrapper removes that packaged local-degree source from the public surface: it takes the three weighted determinant component identifications, the q-normalized theta/log-volume identity, the Theorem 1.10 formula-gap valuation-ball source, and the local arithmetic-degree residual payload, then constructs the local-degree source internally before proving the same two outputs. The p-adic unit-ball Step (xi) synchronization source now projects directly to that local arithmetic-degree residual payload: the Haar defect is the p-adic unit-ball index $p_v^{[K_v:\mathbb{Q}_{p_v}]}$, and Lean obtains the generic residual endpoint by composing the p-adic Haar-index residual, pointwise Haar-defect residual, and summed local arithmetic-degree residual constructors. Thus, on this slice, the residual package is no longer the primitive mathematical assumption; the local Step (xi) synchronization is now the next boundary to lower. This synchronization has now also been lowered to a p-adic arithmetic formula matching source. Instead of taking the synchronized upper-contribution identity directly, the new constructor takes the two local equalities $\mathrm{StepXI}_v = a_v(D_v + C_v)$ and $E_v = p_v^{[K_v:\mathbb{Q}_{p_v}]} + M_v$. Since the local arithmetic upper contribution is definitionally $a_v(D_v + C_v) + E_v$, Lean reconstructs the p-adic Step (xi) synchronization and the expanded Haar-residual identity from these formula pieces. The new bridge from the formula-gap matched valuation-ball arithmetic-degree/p-adic source maps the existing source-backed formula tower into this canonical constructor, so the remaining local-estimate obligation is now the underlying determinant calibration and prime-error split, not a separately supplied Step (xi) synchronization record. The calculation is now exposed directly as well: Lean proves that the local defined residual is exactly the p-adic unit-ball index and then sums these identities to obtain the total residual Haar lower bound used by the C_Θ comparison. At the p-adic unit-ball/local-analytic boundary, Lean now also proves endpoint contracts for the two projected sources: the valuation-ball local-analytic localized source and the expanded p-adic Haar-residual source are both reconstructed from the p-adic unit-ball localized source and the matching local arithmetic-evaluation ledger. The localized source itself also proves the same residual arithmetic directly: its local defined residual is the p-adic unit-ball Haar index, and summing these identities recovers the residual Haar lower bound at this boundary. A matching weakened-boundary countermodel shows why this is not a cosmetic replacement: if the synchronized index is allowed to be arbitrary, the Step (xi) identity and residual/index equality can both hold with zero residual, so the p-adic normalized-place lower law is the source-backed input that rules out the failure. The same lowered source now carries the relative nonvacuity audit and the public-boundary inventory audit directly: Lean constructs the old local-analytic source internally before proving that the q-normalized route, localized C_Θ audit, signed bound, and dichotomy are all obtained without extra public localized-C_Θ, raw signed-bound, hull, or calibration inputs. The same evidence is exposed at the named local-degree public endpoint by endpoint-level route and public-boundary audits, so downstream uses of the public theorem do not need to open the same-index source namespace or invoke the older local-analytic public name to recover the constructed q-normalized handoff, localized C_Θ audit, signed IUT IV bound, relative nonvacuity witness, and dichotomy. The public-boundary inventory records that localized C_Θ, raw signed comparison, principal hull, theta-region principal source, measure calibration, and summand calibration are not separate public inputs on this route. The remaining boundary is therefore the construction of the underlying same-index Hodge-theater/Frobenioid/log-Kummer source package. A further source-hypothesis refinement removes the direct side-condition package from this public boundary: the route now consumes source-labeled q-pilot positivity and normalization hypotheses and constructs the Stage 1 side-condition record internally. The case-bounded same-index boundary now has the corresponding source-obligation-gap form. Its public dichotomy and $C_\Theta \geq -1$ endpoints project the q-pilot positivity and normalization from ‘IUTStage1SourceObligationGap.sideConditionHypotheses‘ before entering the valuation-ball case split. Thus the strongest case-bounded public surface no longer names an anonymous side-hypothesis record, although the paper-level derivation of the source-obligation gap itself remains part of the lower Theorem 3.11 construction boundary. The same public surface has also been lowered through the concrete packet: Lean constructs that source-obligation gap from ‘packet.primitiveConstructor.constructedBundle.toStructuredInputsWithSHE‘ and the Stage 1 side conditions before invoking the case-bounded signed and $C_\Theta \geq -1$ endpoints. Thus the case-bounded route no longer requires an assembled source-obligation gap as a public input at this boundary; the remaining sign/normalization input is the Stage 1 side-condition package. A source-backed specialization now projects those side conditions from ‘IUTStage1SourceHullDetObligations‘: the public signed and $C_\Theta \geq -1$ case-bounded endpoints still build the source-obligation gap from the primitive constructed-SHE bundle, but they read q-pilot positivity and normalization from the hull/determinant obligation object. The remaining lower boundary for this route is therefore the construction of the source hull/determinant obligation package from the paper-level possible-image and hull data. This boundary has now been lowered once more for the same case-bounded surface: the constructed Remark 3.9.5 holomorphic hull/determinant source projects its Theorem 3.11 hull/determinant constructor and hence its ‘IUTStage1SourceHullDetObligations‘; the public signed and $C_\Theta \geq -1$ endpoints consume that constructed source directly. The same endpoint has a stricter principal-product HDD variant: the public route may now consume the Theorem 3.11/Remark 3.9.5 principal-product HDD source and read the source-obligation gap from that source's ‘sourceGap‘ field. A further internal-principal-HDD variant reconstructs this principal HDD source from the compact-open theta-region, the pointwise determinant formula calibration, and the source side hypotheses before entering the same case-bounded endpoint. The bundled same-index case-bounded source is now constructible from the same-index formula-gap source together with the explicit valuation-ball Theorem 1.10 case residual source: determinant synchronization, theta equality, and the formula-gap ledger are read from the formula-gap source, while the distinguished, nondistinguished, archimedean, and distinguished +1 residual comparisons are supplied by the local case source. The same constructor boundary has now been lowered below the formula-gap ledger by ofLocalAnalyticIUTIVSourceCaseBoundedResidualSource: determinant synchronization and theta equality remain explicit at this stage, but Lean constructs the formula-gap matched Theorem 1.10 source from the valuation-ball additive-Haar local-analytic arithmetic-divisor source before assembling the bundled case source. At the localized C_Θ layer the corresponding case estimates can now be supplied directly over the local-analytic valuation-ball source, and the same lowering has now been composed through the same-index case-bounded constructor ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource. Thus the bundled same-index source can be assembled from determinant synchronization, theta equality, the valuation-ball local-analytic IUT IV source, and case inequalities stated over that same source; Lean constructs the formula-gap case residual internally before entering the older same-index case-bounded route. The signed endpoint wrapper dichotomy_ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource now composes this constructor directly into the internal-principal-HDD Corollary 3.12 route, so the signed public boundary no longer needs the formula-gap case source on this branch. Its companion cTheta_ge_neg_one_ofLocalAnalyticIUTIVSourceLocalAnalyticCaseSourceResidualSource now proves the same branch's $C_\Theta \geq -1$ lower bound from the identical local-analytic source boundary. The case-bounded source now also has the projection toLocalAnalyticSource: it constructs the older same-index local-analytic source by projecting the local-analytic ledger from the formula-gap valuation-ball source and proving the total Haar residual inequality from the case-bounded pointwise residual source via the local arithmetic-degree residual theorem. Thus the explicit valuation-ball case split is known to imply the local-analytic endpoint route, rather than merely running in parallel to it. The remaining lower boundary has also been composed into signed and $C_\Theta \geq -1$ endpoints: the public route can now consume the formula-gap source and the explicit local case residual source directly, construct the bundled case source internally, reconstruct the principal HDD source internally, and then enter the case-bounded comparison. The scale-synchronized case-bounded route now removes the raw theta equality from this boundary, and the local arithmetic-degree residual constructor removes the separate total Haar residual field from the scale-synchronized local-analytic source. The weighted-determinant refinement now composes this constructor into the signed and $C_\Theta \geq -1$ endpoints: Lean derives the determinant-source equality from the summand, anchor, and positive tensor-power identifications before entering the same local-degree route. The canonical manifest now records the q-normalized local-arithmetic-degree route: the older scale-synchronized case-bounded source is no longer the canonical endpoint. The q-normalized theta/log-volume identity remains a public source obligation, but Lean uses it with q-pilot positivity to derive the canonical C_Θ-scale synchronization internally. The same scale-synchronized branch now has a local-analytic case-source form: the public IUT IV input is the valuation-ball local-analytic Theorem 1.10 source together with the local-analytic case split, and Lean constructs both the formula-gap matched source and the formula-gap case residual internally before invoking the signed and $C_\Theta \geq -1$ endpoints. This has now been lowered to the procession-bound form on the same branch: Lean reconstructs the expanded local case source from the projected Theorem 1.10 procession upper bound and the distinguished +1 comparison. The same endpoint also accepts the procession-comparison source, separating the Step (xi) identification with procession-normalized local log-volume from the upper-semi procession estimate before reconstructing the procession-bound source internally. It now also accepts the processional-estimate source, deriving that upper-semi comparison from the local Step (v), Step (vi), and Step (vii) estimates. What remains is to derive the weighted determinant component synchronizations, the q-normalized theta/log-volume identity, and the local arithmetic-degree residual package from the concrete theta-region/Hodge-calibrated possible image data and the full IUT IV local estimates. The local-upper refinement defines the local Haar defect as the residual between the IUT IV local arithmetic-upper contribution, the localized Step (xi) determinant term, and the main-log summand; the expanded $StepXI_v + Haar_v = a_v(D_v + C_v) + E_v - M_v$ equality is then derived from the definition of the IUT IV local arithmetic-upper contribution. Conversely, the same-index theorem above uses this expanded local identity together with the total Haar lower bound to recover the formula-gap residual inequality. The localized-data constructor then constructs the localized C_Θ source itself from the localized Ob3/Ob4 determinant source, Haar-normalization defects, main-log contributions, and finite-place summation identities before running the same route audit. These variants remove the record Ob3/Ob5 determinant source from five higher source-spine entry points; the remaining explicit inputs at this boundary are the localized family identification, the exact theta/family-hull calibration, and the hull/determinant obligations synchronization. There is now a stricter constructor-built variant of this source-spine route: the rebuilt source-spine theta record is definitional on the Theorem 3.11 image family, and a constructor-built possible-image Step (xi) source supplies the record-canonical hull/determinant bridge, q-region, q-positivity, normalization, Ob3–Ob6 determinant bridge, and the record-bridge Ob7 input without an intervening obligations synchronization. Its product-formula Ob7 variant now takes finite weighted determinant prime-strip data attached to that same constructor-built determinant source; Lean constructs the coric $\Theta^{\times\mu}$ invariant and proves the determinant/global prime-strip equality before entering the source-spine route. The obligations-backed version of this product-formula route now constructs the possible-image constructor-built source at the source spine's selected q-choice; the selected-q equality is therefore definitional on that path, and the remaining endpoint facts are projected from the constructed determinant source, the product-formula Ob7 source, and the single hull/determinant obligation synchronization. A stricter weighted-partition variant lowers this once more: from a prime-strip lift, summand-place map, finite weights summing to one, and local Gaussian volume calibration, Lean builds the determinant product-formula source internally before applying the same source-spine/action-packet route. This weighted-partition route now has the same selected-q, obligations-backed constructor path: the possible-image constructor-built source is assembled at the source spine's selected q-choice, and the endpoint records the constructed summand-place map, conversion ratio, global finite product formula, determinant/global equality, bridge synchronization, positivity, and normalization. A constructor-sourced variant removes the ambient 'IUTStage1SourceHullDetObligations' object from this selected weighted-partition path: it builds the selected possible-image constructor directly from the record-canonical bridge equality, q-positivity, and normalization, and proves that the obligations visible at the endpoint are definitionally the obligations produced by the constructed Theorem 3.11 hull/determinant constructor. The bridge equality and side conditions are still explicit lower inputs on this variant. The exact-theta selected Step (xi) paper-trace constructor now also accepts this symmetry-label transport source directly and projects the older action-packet transport internally; the source-spine record constructor has the same symmetry-label entry point. The packet-facing completion route can now construct the packet-indexed symmetry-label wrapper from the possible-image witness and explicit symmetry-label, upper-semi, and procession transports, so that wrapper is no longer forced to appear as a public theorem argument at this surface. At the public paper-trace route boundary, the remaining payload audit has now been split so the caller supplies only the non-Step-(xi) payload; the legacy Step-(xi) audit fields are derived internally from the structured Step-(xi) source. A stricter concrete-localized additive-Haar wrapper now makes this split explicit at the normalized Haar-modulus boundary. This has now been tightened again: the public wrapper may take only the seven lower arithmetic/formula fields from Theorem 1.10/IUT IV, and a further wrapper derives those seven fields from the IUT IV C_Θ source ledger itself. The Step (x) realified-packet field is projected from the vertical log-Kummer package, the IPL/SHE/APT fields are projected from the Theorem 3.11 bridge package, and the Step (xi) arithmetic-degree, localized determinant, adjusted log-volume, and endpoint payload slots are constructed from the normalized finite-summand source-route audit, which explicitly carries the projected additive-Haar, Haar-modulus, and localized-adjusted Hodge-summand expansion boundaries. In the newest normalized additive-Haar route, local additive-Haar normalization, the Theorem 1.10 arithmetic-divisor split, the finite-place Haar defect, the arithmetic-gap estimate, and the local-to-global C_Θ comparison are read from IUTIVCThetaObligations, not supplied as a separate residual record. When a constructor-built Step (xi) hull/determinant source is present and its canonical scale is identified with the IUT IV finite-place scale, the residual endpoint is strengthened further: the final C_Θ comparison is threaded through the constructor-built Step (xi)/IUT IV local-to-global audit rather than through the generic ledger proposition alone. The newer constructor-coupled arithmetic source goes one layer lower: it constructs the IUT IV finite-place ledger with the constructor-built canonical scale as the finite-place canonical scale, so the public handoff no longer asks for a separate scale equality. Its remaining numerical input is the lower arithmetic gap inequality $C_{\Theta,\mathrm{can}} + 1 \leq \mathrm{upper} - \mathrm{main}$. The finite-place local-to-global constructor now lowers this numerical input: local canonical scales, arithmetic/log contributions, and Haar normalization defects are summed to derive the arithmetic gap before the coupled IUT IV ledger is constructed. This finite-place source now has its own endpoint, and the localized C_Θ handoff records that endpoint explicitly before projecting the final signed comparison. The localized local-term constructor lowers this one step further: the local canonical scales are the weighted adjusted log-volumes of the localized Ob3/Ob4 determinant source, and each local upper term is decomposed as Step (xi) log-volume plus Haar defect plus main-log contribution. The arithmetic-divisor-backed additive-Haar source now projects the reusable IUT IV local-analytic and Theorem 1.10 C_Θ data from its matched local-degree, formula-gap, and local-global audits; the projection itself now factors through the typed additive-Haar formula-gap arithmetic-divisor source, so the Proposition 1.4/1.5 local endpoints, formula-to-gap comparisons, and local different/conductor nonnegativity bounds are read from that lower source. The strongest paper-trace audit now uses the same projections instead of rebuilding those C_Θ records inline. After the constructor-built determinant is synchronized with the localized Step (xi) determinant, this source constructs the localized local-term C_Θ source and derives the residual-payload audit, the constructor-built arithmetic gap, and the final C_Θ comparison. The same arithmetic-divisor-backed matched local-degree source now also projects into the canonical p-adic unit-ball arithmetic formula-matching boundary. Thus the p-adic Step (xi) synchronization audit can be reached from the Theorem 1.10 arithmetic-divisor local-degree branch, rather than only from the formula-gap arithmetic-degree/p-adic bridge. This stronger residual handoff now also feeds the base normalized and non-normalized additive-Haar public route constructors, so those boundaries can be instantiated from the constructor-built Step (xi) source instead of reverting to the generic IUT IV residual. The same constructor-built handoff now lifts through the intrinsic finite-extension Haar-modulus normalized and finite-summand additive-Haar routes, and through the corresponding unit-ball Haar-character normalized and finite-summand wrappers. It now also reaches the residue-module/unit-ball normalized and finite-summand wrappers; the normalized residue-module route has the same constructor-coupled no-standalone-scale- equality endpoint. This CTheta-derived arithmetic boundary has also been lifted through the intrinsic finite-extension, unit-ball Haar-character, and residue-module/unit-ball normalized public routes, so those stronger p-adic endpoints can now use the same source-derived arithmetic payload. The direct non-normalized additive-Haar route also has finite-place and localized constructor-built C_Θ entry points, so it no longer has to pass through the finite-summand wrapper merely to consume localized Step (xi) local-term data. The unit-ball normalized wrapper has its own full-boundary audit, and the strongest residue-module/unit-ball wrapper now factors through it before projecting to the intrinsic Haar-modulus normalized route. Its audit combines the Step (xi) residue-module source audit, the transported residue-module factor handoff, the projected unit-ball normalized audit, the IUT IV arithmetic-residual audit, and the definitional equality with the projected normalized public route. The parallel non-normalized unit-ball Haar-character and residue-module/unit-ball Haar-character finite-summand routes now have the same CTheta-derived arithmetic handoff: their public wrappers project to the intrinsic Haar-modulus/additive-Haar finite-summand route, and their full audits record the Step (xi) source boundary, the projected finite-summand boundary, the IUT IV residual audit, and the resulting remaining arithmetic-payload audit. On the hull/determinant side, the Ob3/Ob5 compatibility record can now be constructed from the lower equality $\mu_{\log}(\varphi(P_B)) = \widehat{\deg}_{\mathrm{norm}}$: the concrete quotient theta-pilot constructor builds the canonical hull approximant versus weighted-determinant compatibility internally from that equality. The exact-theta variant also uses $\theta_{\mathrm{signed}} = \mu_{\log}(\varphi(P_B))$ to derive the Ob4 tensor-power bound, so that bound is no longer a separate input on this constructor path. The constructor-built possible-image Step (xi) source now has the same lowered entry point: from a Remark 3.9.5 hull operator, adjusted Ob3/Ob4 determinant data, and these two log-volume equalities, Lean reconstructs the Ob3/Ob5 compatibility and tensor-power bound internally before entering the record-canonical hull/tensor bridge. The minimal-hull-system and product-hull variants expose this path without a prebuilt hull shadow; the bridge equality, measure synchronization, q-positivity, and normalization remain explicit lower inputs. The product-hull exact-theta endpoint now packages the source endpoint, product-hull provenance, reconstructed compatibility, derived tensor-power bound, selected-product-hull containment, and the resulting signed comparison as checked consequences of that lowered constructor. An obligations-backed variant lowers the remaining side-condition surface: bridge equality, q-pilot positivity, and source normalization are projected from a single `IUTStage1SourceHullDetObligations` object. The explicit synchronization left at this boundary is now the equality between that obligations bridge datum and the record-canonical product-hull/tensor-power datum. The concrete-localized route wrapper takes the minimal-hull source, localized vector-bundle determinant decomposition, and $\Theta^{\times\mu}$ Ob7 source directly, constructs the paper-derived Step (xi) source internally, and then invokes the preferred route. A stronger hull-side wrapper now goes below the minimal-hull argument: it constructs the concrete hull source internally from the source-spine equality-quotient possible-image family and the principal $\lambda \cdot O$ hull source, then threads that hull source through the localized determinant and Ob7 route. This principal-product route now has a strict variant in which the determinant upper bound is derived from Θ_{signed} equal to the localized family-hull log-volume. The strict principal-product localized wrapper has now been promoted to the named public Step (xi) paper-source boundary. The older route theorems still exist as compatibility layers, but the advertised public audit no longer takes an already constructed Step (xi) source. A product-formula strengthening of this public audit now also constructs the localized $\Theta^{\times\mu}$ Ob7 source from the localized determinant source, local Gaussian summand calibration, finite determinant product formula, and coric unit-character data. Thus the public route can consume product-formula data in place of a prebuilt Ob7 compatibility record. At this localized Ob7 constructor boundary, the tensor-power upper bound is derived from $\Theta_{\mathrm{signed}} = \mu_{\log}(\varphi(P_B))$ and the localized determinant comparison, rather than supplied beside the product-formula source. There is now a determinant-calibrated public product-formula theorem whose audit boundary uses this constructor directly; the older theorem remains as a compatibility layer for routes not yet migrated. The downstream raw-adjusted, common-place, diagonal restriction, and environment-theta public wrappers now use the same constructor, so the Step (xi-a)–(xi-g)/Ob7 trace is no longer backed by a separate inline tensor-power estimate along this wrapper ladder. The public audit now has a stricter raw-adjusted variant: it consumes raw adjusted-localization Gaussian calibration data, constructs the localized product-formula source internally, and then invokes the product-formula public route. Each determinant summand is assigned a Gaussian place and raw ratio, the partition weight is computed from the Ob3 localization weight times this ratio, and Lean derives the weighted summand calibration and global product-formula equality from the Gaussian local-global Frobenioid restriction identity. This has now been lowered once more: the public route also accepts a globally normalized raw-adjusted source in which the raw ratio is computed as adjusted raw log-volume divided by the nonzero Gaussian global log-volume. The Ob3 weighted determinant sum together with determinant/global prime-strip equality proves the finite normalization. The Ob7 prime-strip boundary has also been lowered: common environment and Gaussian prime-place coordinates plus one valuation-indexed coric character now construct the prime evaluation, $\Theta^{\times\mu}$ lift, and coric invariant internally. A further public wrapper constructs the local-global collection, its environment anchor, and the local environment/Gaussian evaluation from global-to-local restrictions and local object identifiers; the local realified objects are forced by the restricted global prime log-volumes. In the diagonal finite valuation-index route, the environment and Gaussian prime-place equivalences are identity maps constructed internally. The preferred non-vacuous public diagonal wrapper now uses the additive-Haar compact-open factor boundary over the principal product hull system, together with an audited Hodge–Arakelov theta evaluation, global-to-local restriction data calibrated at the Hodge canonical full-label Gaussian degree, and the scalar equality between the Hodge theta-monoid degree and the localized adjusted determinant sum. The environment/Gaussian realified Frobenioid evaluation is constructed internally from zero-divisor ordinary/theta/log-shell copies whose unit log-volume is the Hodge canonical degree; normalization proves that the Gaussian theta divisor log-volume is this Hodge canonical degree. The Hodge evaluation identifies the canonical degree with the theta-monoid degree. The finite Hodge summands are no longer supplied as a public family: they are fixed internally to be the localized weighted adjusted determinant contributions, and the scalar equality proves that their finite sum is the theta-monoid degree. The structure-sheaf log-volume is identified with a distinguished direct summand of each localized arithmetic vector bundle. Each direct-summand log-volume is projected from the normalized log-volume of a local compact-open tensor factor constructed from additive Haar data. The additive-Haar layer now has a strict factor constructor: from the source-level inequality $1 < \mu(U)_{\mathbb{R}}$ for the selected non-structure compact open, Lean proves positivity of $\log(\mu(U)_{\mathbb{R}})/[K : \mathbb{Q}_p]$, then uses the norm-square projection to derive the nonzero direct-summand norm. This generic interface has been instantiated by a concrete finite-extension local-field source: the selected factor is the inverse base-prime dilation $p^{-1}O_K$, whose compact-openness and Haar mass $p^{[K:\mathbb{Q}_p]} > 1$ are proved in the source core and threaded through the preferred additive-Haar public wrapper. The same wrapper can now consume a family-hull-indexed calibrated localized hull-cover source, deriving the old family-hull containment, log-volume summation, and pointwise localized-region determinant identities from the index-fiber presentation, finite additivity, and local vector-bundle calibrations. A stricter variant lowers the volume-summation input from a global finite-additive hull law to a cover-local log-volume measure model for the localized regions under construction. The newest direct-product variant also constructs the family-hull index from local factor regions: the direct-product cover identity gives existence of an index, and local factor separation proves the required fiber equality. A tensor-measure refinement constructs the cover-local measure model from direct-product cell log-volumes and the tensor-cell summation formula, so the direct-product public route no longer needs an independently supplied finite-cover measure model. A source-core refinement lowers this again: the tensor cells are now the localized determinant summands from the additive-Haar source-core decomposition, and the tensor-cell finite-sum formula is derived from the source-core localized log-volume theorem together with the direct-product cover identity. The remaining input at this boundary is the geometric identification of the localized regions with the direct-product local-factor cells and their cover of the principal-product family hull. A charted local-ring refinement lowers the factor interface itself: the local factors are no longer an arbitrary region family, but the regions of per-place local-ring/vector-bundle calibration objects whose chart log-volumes are identified with the corresponding direct-summand log-volumes. The cover identity and factor separation are still explicit geometric inputs. A valuation-ball charted refinement now lowers the factor calibration one step further: each local-ring chart is projected from a valuation-ball/additive-Haar factor calibration, and Lean derives the charted local-ring cover interface from the valuation-ball regions, their separation, and the valuation-ball family-hull identity. The p-adic finite-extension branch now has the same principal valuation-ball lowering: its direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover before entering the p-adic charted route. The p-adic branch also has principal-hull and hull-formation alignment refinements: the valuation-cover/public-hull equality is derived from the principal valuation-ball hull theorem, and the source-core and valuation-cover possible-region equalities are projected from the concrete `principalProductHullFormationData` family. The p-adic calibration boundary is also lowered: the localized calibration and per-factor valuation-ball calibration are transported from the principal valuation cover across the principal-hull equality, so they are no longer supplied independently at this level. The additive-Haar principal valuation-ball refinement lowers this again: the direct-product cell identity, local-factor separation, and family-hull equality are projected from the SourceCore principal valuation-ball product-hull cover. The selected valuation-ball branch now also transports exact possible-image $\Xi(P_B)$ witnesses across carrier equivalences: Lean proves this first for the holomorphic hull operator and possible-image family, then applies it to the radius-cover exact-Ξ source. A public adapter now aligns this transported source with the Theorem 3.11 quotient family using quotient-index matching, pointwise possible-region equality, and hull-operator equality; the public exact-Ξ log-volume calibration is then derived rather than supplied. The quotient-index matching has also been lowered to concrete choice-level data: a selected valuation-ball index attached to each Hodge-theater/log-theta choice, invariance under the Theorem 3.11 equality quotient, and representatives over selected indices construct the quotient equivalence used by the adapter. Thus the local selected-q Ob5–Ob6 boundary now constructs the canonical $\Phi(P_B)$ family and the exact $\Xi(P_B)$ family from this calibration before the global C_Θ comparison; closed-family and product-hull wrappers feed the same audit. The remaining boundary is to keep lowering the additive-Haar source-core localized determinant decomposition. A principal-hull-aligned refinement removes the direct valuation-cover/public-hull equality from this boundary: Lean derives it from the lower principal valuation-ball cover's own principal-hull theorem and the equality between that principal hull system and the public Theorem 3.11 principal-product hull system. A hull-formation possible-region refinement lowers the remaining possible-region equality: the source-core and valuation-cover families are aligned with the concrete family inside ‘principalProductHullFormationData‘, and Lean derives the public ‘principalProductPossibleRegion‘ equality by projection. The p-adic source-core localized determinant alignment is now constructed from these transported valuation-cover fibers: the finite-extension source core is built from transported local calibration data and finite-extension unit-ball Haar factors, and Lean proves the projected vector-bundle and localized-region equalities consumed by the older route. The remaining lower matching datum is the factor-level equality between each projected finite-extension valuation ball and the transported valuation-cover factor. A finite-extension-backed valuation-cover wrapper now lowers the valuation-cover side itself: the lower principal valuation cover is projected from finite-extension p-adic Haar data before it enters the constructed source-core route. A hull-formation-selected intrinsic cover constructor fixes the lower p-adic cover's hull system to the public principal-product hull system and fixes its possible-region family to `principalProductHullFormationData`; the intrinsic possible-region comparison is therefore produced by construction, while the principal-hull comparison is exposed as the lower equality between the p-adic principal hull source and the public principal-product hull system. A public-principal specialization now constructs this lower p-adic principal source as the `ULift` copy of the public principal-product source and proves the product-hull intersection law by transport through the public intersection theorem. Lean now also proves that the induced holomorphic hull system is unchanged by this `ULift`: the proof reduces equality of `IUTStage1Remark395HolomorphicHullSystem` records to equality of the `isHull` predicate and `logVolume` field via the generated constructor-injectivity theorem. The intrinsic wrapper also removes the separate public finite-extension factor family: Lean transports the lower cover's finite-extension factors across the same hull-system equality used for the valuation-cover calibration fiber and derives the old factor-matching field internally. An inverse-dilation refinement lowers the analytic input further: each transported valuation-ball Haar factor is identified with the inverse base-prime unit ball of a constructed finite-extension p-adic Haar source, and Lean derives $1 \leq \mu(U)_{\mathbb{R}}$ from the strict mass theorem $1 < \mu(p^{-1}O_K)_{\mathbb{R}}$. The strict selected determinant summand is now tied to this same inverse-dilation family: the unified public route projects the selected summand from the ($\texttt{positiveIndex}, \texttt{positiveSummand}$) member of the valuation-cover family, rather than accepting an independent selected-factor inverse-dilation source and equality. A boundary audit theorem now exposes this dependency shape by proving both the all-factor and selected-summand inverse-base-prime unit-ball equalities from the same family. The hardened preferred wrapper lowers the theta scalar input one more step: it takes the Hodge/family-hull equality and derives the localized adjusted determinant-sum equality by the localized hull decomposition theorem before invoking the unified inverse-dilation route. This wrapper has now been lowered again: the inverse-dilation branch can start from the finite localized Hodge-summand formula and derive the Hodge/family-hull comparison from the same Remark 3.9.5 decomposition. Lean also proves the converse source-facing expansion needed by the public route: a localized-adjusted inverse-dilation source projects to the finite Hodge-summand source by unfolding the localized adjusted sum as the finite sum of weighted adjusted log-volumes. This expansion is now also available from the constructed Gaussian/Hodge inverse-dilation wrapper itself: Lean reads the constructed Hodge theta-summand family, proves its pointwise equality with the localized weighted adjusted log-volumes, and then constructs the finite Hodge-summand calibrated source directly. The corresponding public finite-summand route now enters through this constructed summand-calibrated source, rather than first projecting to the older localized-adjusted wrapper. The Haar-modulus-backed inverse-dilation wrapper now exposes the same finite-summand expansion boundary after projecting its measure-level inverse-base-prime cover to the generic inverse-dilation source; its audit records the direct constructed Hodge-summand boundary, not the older localized-adjusted expansion detour. The additive-Haar construction layer now uses this same route after it constructs the Gaussian/Hodge localized-adjusted source from compact-open factors and the strict inverse-base-prime mass theorem. The normalized structure-sheaf branch now uses the finite-summand route as well: after deriving the direct-summand handoff from compact-open normalized Haar data, the residual and IUT-IV public wrappers enter through the Hodge-summand expansion endpoint rather than the older family-hull endpoint, and the additive-Haar arithmetic-degree payload is certified by the same normalized finite-summand audit. The non-normalized additive-Haar/Gaussian-Hodge companion now has the parallel summand audit and an IUT-IV arithmetic-source wrapper for the finite-summand endpoint, so its determinant/calibration payload is no longer tied only to the older family-hull route. The residue inverse-base-prime normalized finite-summand branch now uses this same direct construction in its IUT IV and constructor-built arithmetic wrappers: Lean builds the normalized additive-Haar arithmetic obligations from the finite-summand audit instead of routing these wrappers through the older direct-summand public endpoint. The same finite-summand endpoint has now been lifted through the intrinsic finite-extension Haar-modulus, unit-ball Haar-character, and residue-module quotient wrappers, so those local arithmetic sources no longer expose only the family-hull route at their public boundary. On the normalized branch, the unit-ball Haar-character wrapper now has its own CTheta-derived endpoint and full boundary audit, and the residue-module normalized route now factors through that unit-ball audit before the intrinsic Haar-modulus normalized boundary. On the non-normalized branch, the intrinsic finite-extension Haar-modulus wrapper now constructs the same p-adic arithmetic-degree obligations directly from its projected additive-Haar source. The non-normalized unit-ball Haar-character wrapper and the residue-module/unit-ball Haar-character wrapper now also have CTheta-derived finite-summand endpoints and full boundary audits, so their remaining additive-Haar arithmetic payload is projected from the IUT IV ledger rather than supplied separately; these endpoints now build the p-adic arithmetic-degree obligations at their own projected finite-summand boundary instead of delegating that construction to a lower generic Haar-modulus wrapper. The finite-place local-global and localized Step (xi) C_Θ wrappers now use their own residual payload constructors at the same boundary, both for the non-normalized finite-summand endpoints and for the normalized structure-sheaf/direct-summand endpoints, instead of first collapsing to the coupled IUT-IV arithmetic-source projection. It now also lowers the structure-sheaf input: the hardened wrapper takes localized direct-summand calibration and derives the additive-Haar normalized compact-open equality by the additive-Haar projection theorem. A complementary non-vacuous Haar-modulus wrapper now uses the opposite, source-level direction required for the additive-Haar construction: it takes the localized structure-sheaf equality with the normalized compact-open Haar factor and derives the Step (xi) structure-sheaf/direct-summand equality from the additive-Haar localized vector-bundle decomposition theorem. The current preferred wrapper lowers the surrounding Gaussian/Hodge boundary again: the restriction local-global object, ordinary-log-volume/Hodge-degree calibration, direct-summand data, and Hodge/family-hull equality are projected from the constructed Gaussian/Hodge localized-adjusted source. It now lowers that source boundary further: Lean constructs the Gaussian/Hodge localized Ob7 source from additive-Haar compact-open factors, derives the selected strict mass from the inverse base-prime factor, projects the Haar-modulus valuation-cover side from the intrinsic finite-extension valuation-cover source, and derives the inverse-dilation Haar-modulus family from unit-ball Haar-character data. The latter is now itself constructed from residue-module quotient data: O_v/p_vO_v, $\#\kappa_v = p_v$, and $\dim_{\kappa_v}(O_v/p_vO_v) = [K_v : \mathbb{Q}_p]$ give the coset decomposition and $\mu(p_vO_v) = p_v^{-[K_v:\mathbb{Q}_p]}$. The normalized-Haar direct-summand handoff has now been propagated through this intrinsic finite-extension and residue-module route as well: the residue source supplies the structure-sheaf equality with the normalized additive-Haar compact-open factor, and Lean derives the Step (xi) structure-sheaf/direct-summand equality before invoking the non-vacuous comparison route. The transported-factor handoff audit follows the residue-module source into the localized additive-Haar vector-bundle factor: the transported valuation-ball factor and the localized determinant factor are both identified with the inverse base-prime unit ball $p^{-1}O_K$ constructed from the residue quotient data. Lean now also projects this equality to the concrete scale invariants used downstream: Haar mass $p^{[K:\mathbb{Q}_p]}$ and strictly positive normalized log-volume for both factors. This handoff is threaded through the additive-Haar residue audit, the normalized residue audit, and the strongest CTheta full-boundary audit, so the public audit object exposes the determinant-factor match directly. The equality boundary has also been lowered in two stages. A component-matched residue source asks for equality of the defining additive-Haar compact-open data fields, and Lean reconstructs the previous full source equality by additive-Haar extensionality. Below that, a scale-component residue source matches the transported factor directly against the finite-extension inverse-dilation mass source: local ring, residue prime, finite degree, realization, Haar measure, $p^{-1}O_K$, and base-prime dilation. A further valuation-ball source now lowers this once more: the transported local factor is matched before additive-Haar projection, including its valuation norm, selected valuation ball $\{x \mid v(x) \leq r\} = p^{-1}O_K$, Haar measure, local realization, and pointwise base-prime dilation. Lean derives the previous scale-component source from this valuation-ball match, and then projects through the unit-ball Haar-character and intrinsic Haar-modulus wrappers, so the lower valuation ladder consumes the same concrete inverse-dilation components rather than a fresh equality at each layer. The valuation-ball source itself has also been lowered into SourceCore: from the constructed finite-extension dilation-mass source and a positive-radius identification $p^{-1}O_K = \{x \mid v(x) \leq r\}$, Lean constructs the valuation-ball additive-Haar source and proves the inverse base-prime component match. Thus this part of the interface no longer needs to be entered as a single opaque equality of normalization records. Under the normed-algebra compatibility for a finite extension $K/\mathbb{Q}_p$, the radius identity itself is now derived: Lean proves from $\|\mathrm{algebraMap}(p)\| = \|p\|_p = p^{-1}$ that the inverse base-prime unit ball is exactly the valuation ball of radius p, and constructs the corresponding inverse base-prime valuation-ball source. This radius computation also exposed a hard obstruction in the surrounding Step (xi) interface: if the transported intrinsic finite-extension factor is still the valuation-unit-ball cover, then its selected radius is 1, while the constructed inverse base-prime factor has radius p. Lean proves that the corresponding normed finite-extension match source is locally contradictory, via `NormedFiniteExtensionInverseBasePrimeValuationBallMatchedSource.no_unitBallCover_factor`. The replacement local cover has now been added: SourceCore constructs a finite-extension inverse-base-prime valuation-cover packet whose local selected factors have radius p from the outset, projects it to the generic valuation-ball tensor packet, and proves the inverse-base-prime component match for every local factor. Step (xi) also has a hull-formation-selected public principal-product sibling route for this radius-p cover, so the unit-ball branch and the inverse-dilated branch are separated explicitly. The residue-module handoff has now been moved onto this inverse cover at the local-factor level: the new handoff identifies the cover's inverse-dilation mass source with the residue-module constructed mass source and derives the equality between the radius-p valuation factor's additive-Haar normalization and the residue-module inverse-base-prime unit-ball normalization. The preferred-public normalized additive-Haar route has now been migrated to this inverse-cover projection as well: its Haar-modulus valuation-cover argument is definitionally the radius-p residue-module inverse-cover source, rather than the legacy unit-ball intrinsic cover. The non-normalized additive-Haar/Gaussian-Hodge route has now been added for the same inverse-cover source: the residue-module wrapper supplies the localized structure-sheaf/direct-summand comparison, projects the Haar-modulus valuation-cover argument from the radius-p inverse cover, and then reuses the generic additive-Haar compact-open ladder to construct the Gaussian/Hodge localized-adjusted Ob7 source. Thus the normalized public wrapper is no longer the only residue-module inverse-cover entry point. The additive-Haar arithmetic-degree obligation package for this non-normalized route is now constructed as well: its Step (xi) determinant/calibration slots are read from the additive-Haar/Gaussian-Hodge finite-summand source audit, and the remaining lower arithmetic residual is projected from the IUT IV C_Θ ledger. The same non-normalized finite-summand route now also has constructor-coupled, finite-place, and localized Step (xi) C_Θ entry points through the intrinsic, unit-ball Haar-character, and residue-module/unit-ball finite-extension wrappers. This finite-summand endpoint is now exposed for the residue inverse-base-prime wrapper itself: its audit records the radius-p residue inverse-cover projection together with the inherited Haar-modulus/additive-Haar finite-summand expansion boundary. The non-normalized residue inverse-cover branch now also has constructor-built, coupled, finite-place local-global, and localized Step (xi) C_Θ handoffs, so this arithmetic layer no longer has to pass through the looser localized-adjusted/family-hull endpoint. The normalized residue inverse-base-prime wrapper reaches this finite-summand endpoint as well. Lean reconstructs the structure-sheaf/direct-summand handoff from the normalized compact-open Haar structure-sheaf equality and then projects to the generic normalized Hodge-summand expansion route. Constructor-built, coupled, finite-place, and localized Step (xi) C_Θ sources now thread this normalized residue endpoint without separately supplying local canonical scales; the intrinsic and unit-ball Haar-character normalized parent wrappers expose the same constructor entry points. This boundary has been lowered once more: the residue inverse-cover route can now take a constructed Gaussian/Hodge localized-adjusted source over the same residue-derived localized hull vector bundle, and Lean projects from it the Hodge local-global comparison, selected direct-summand data, structure-sheaf equality, and localized-adjusted theta equality required by the older additive-Haar route. The same residue corridor now has a summand-form source: the localized-adjusted theta equality is derived by summing the finite Hodge summands and identifying them pointwise with localized weighted-adjusted determinant terms. A companion restriction-local-global wrapper now derives the older public source's global object, localization, local object identifiers, and pointwise Hodge/local-prime equality from the environment-anchored realified Frobenioid restriction source. The constructed Gaussian/Hodge residue source now also factors through this restriction wrapper. The summand-form residue source factors through the same layer as well: its audit combines the finite-summand theta expansion with the projected local-global restriction boundary, so the anchored local-global source is produced by the constructed summand source rather than bypassed by the older public fields. The inverse-dilation principal-product branch now lowers this summation boundary again: a square-weight source derives the localized-adjusted theta equality from normalized finite Hodge weights together with pointwise identities between each weighted Hodge contribution and the corresponding localized weighted-adjusted determinant term. A stricter normalized-summand source now derives these weights from the constructed Hodge summands by dividing by the nonzero theta-monoid degree; Lean proves the resulting weights have total 1 and projects to the square-weight route. A raw-positive variant derives the required theta nonvanishing from local determinant positivity, and a direct-summand refinement now derives that raw positivity from the Ob3 comparison: the structure sheaf is one nonnegative direct summand, while a distinct summand is strictly positive. The square-weight branch now also projects through this direct-summand raw-positive route, so the public square presentation and the positivity normalization are no longer parallel route-level assumptions. The Haar-modulus inverse-dilation and additive-Haar public branches have now been promoted to the same strict route: their compact-open/valuation-cover constructors feed the direct-summand raw-positive normalization instead of stopping at localized adjustment. The intrinsic p-adic finite-extension, unit-ball Haar-character, and residue-module/unit-ball wrappers now expose the same strict projection, so these lower arithmetic branches no longer land only at the older localized-adjusted finite-summand endpoint. What remains below this route is to construct the underlying local square/direct-summand determinant factors and the pointwise determinant identities from the lower Hodge-theater and log-theta-lattice data. The full-boundary audit for the current route also ties the IUT IV arithmetic residual to the same public wrapper. The remaining principal-hull boundary is now lower than the public route wrapper: the coordinate and real-line public endpoints construct the principal source and paper trace internally, and the coordinate scalar-image landing/preimage laws are derived from the exact-image formula. The remaining obligation is to construct that exact local image formula from the lower Hodge-theater/log-theta-lattice data. The selected valuation-ball exact-Ξ audit now consumes the same coordinate scalar-image source, rather than a parallel selected scalar wrapper. For the selected valuation-ball hull, the nonzero-scalar multiplication route now constructs the selected scalar-parameter source and its Ob3/Ob5 handoff; the remaining work is to construct the lower transported-factor/residue-module matching data directly from the same Hodge-theater/log-theta local-field package. The valuation-unit-ball finite-extension branch remains only a normalized legacy boundary: the selected radius is forced to be 1, so $\mu(O_v) = 1$ constructs weak nonnegativity but also proves that the normalized log-volume of the selected unit-ball factor is 0. Thus the strict-positive unit-ball route is explicitly quarantined as vacuous, now also at the exact intrinsic finite-extension public endpoint. The p-adic finite norm-square/direct-summand localized-adjusted route has now been checked against the same normalization: Lean proves that the selected unit-ball direct-summand norm is forced to be 0, so the nonzero-norm version is also a quarantined unit-ball branch rather than a non-vacuous determinant route. Its wrappers through the family-hull indexed measure model, direct-product measure model, tensor-measure cover, source-core tensor cover, charted local-ring cover, valuation-ball cover, principal-hull aligned cover, transported valuation-cover calibration, constructed source-core valuation-cover, finite-extension-backed cover, and intrinsic finite-extension cover wrappers remain checked compatibility infrastructure, but not a completion witness. The explicit norm-square compatibility route `preferredPublicConcreteStepXI311312NormSquareRoute` consumes this concrete finite-extension valuation-cover data, the localized-adjusted norm-square/direct-summand source, and the explicitly lower-level additive-Haar remaining-payload audit; Lean constructs the assembled non-Step-(xi) payload internally, so this boundary no longer accepts a preassembled Step (xi) audit. The short preferred route `preferredPublicConcreteStepXI311312Route` has now been moved to the constructor-built localized C_Θ surface: it projects the IUT IV arithmetic residual and additive-Haar payload from the constructor-built possible-image holomorphic-hull/determinant source and the localized Step (xi) finite-place local-term source, rather than accepting those objects independently. The new constructor-built localized C_Θ goal-evidence audit records this short surface directly: it includes the constructor-built Remark 3.9.5 Step (xi) boundary, localized finite-place C_Θ endpoint, local-term-to-IUT-IV C_Θ handoff audit, local-to-global C_Θ audit, and the additive-Haar remaining payload derived from the localized arithmetic source. The residue inverse-base-prime projected Gaussian/Hodge additive-Haar branch, including its finite-Hodge-summand refinement, has also been lowered to this finite-place/localized constructor-built C_Θ surface, so it no longer exposes an independent IUT-IV arithmetic ledger after the summand expansion has been constructed. The same lowering now covers the residue restriction/local-global variants: the base restriction source, the projected Gaussian/Hodge restriction source, and the summand-projected restriction source all enter through the localized constructor-built C_Θ handoff. It also covers the plain direct residue inverse-base-prime additive-Haar branch and its normalized structure-sheaf companion, so those direct routes now use the localized constructor-built C_Θ handoff as well. A stricter restriction-calibrated public variant `preferredPublicConcreteStepXI311312RestrictionCalibratedRoute` now uses the environment-anchored local-global Frobenioid restriction package to derive the Hodge local-prime comparison before entering this same constructor-built localized C_Θ handoff. Its route audit now also carries the inverse-dilation strict-selected audit from the same restriction-calibrated source, so strict Haar mass, positive normalized compact-open log-volume, and the selected nonzero direct-summand norm are exposed at the public restriction-calibrated boundary rather than as a detached side result. A corresponding restriction-calibrated goal-evidence package now records this route audit together with the projected constructor-built localized C_Θ goal evidence. The handoff audit identifies the constructor canonical scale with the finite-place scale, proves the summed Step (xi) Haar-defect arithmetic gap, and derives the global C_Θ bound used by the ordered-real dichotomy. A stricter product-hull localized C_Θ wrapper now constructs the Step (xi) source from the Remark 3.9.5 product-hull system, Ob3/Ob4 adjusted determinant data, and exact theta/family-hull log-volume equalities before entering this short preferred route; its audit carries product-hull provenance together with the localized C_Θ payload derivation. The same audit now exposes the constructor-sourced Step (xi) endpoint, where the obligations object is definitionally the one produced by the constructed Theorem 3.11 hull/determinant constructor. The obligations-backed variant projects the bridge equality, q-pilot positivity, and normalization from a single `IUTStage1SourceHullDetObligations` object, leaving only the identification of that bridge datum with the record-canonical product-hull/tensor-power construction explicit. The stricter canonical-HDD wrapper now moves that synchronization below the preferred route into `ProductHullCanonicalHDDCommonContainerSource`: its endpoint records that the package HDD hull/determinant bridge equals the constructor-generated obligations bridge and hence the record-canonical product-hull/tensor-power bridge, while the preferred route itself consumes this lower source object and the localized C_Θ source. Below that surface, the new `withHDDHullDetBridge` constructor builds a package whose HDD hull/determinant bridge is definitionally the record-canonical product-hull bridge, and the corresponding definitional-package audit verifies preservation of the chart and signed q/Θ endpoints. The Theorem 3.11 source record is now transported across this canonical-HDD package by `canonicalHDDRecord`; the transfer audit records that the transported record carries the constructed coric HDD bridge, preserves the possible-image family, retains the target-union equality, and keeps the separated Hodge-theater histories. The canonical-HDD preferred route now enters the standard short route through a constructor-built Step (xi) source over this transported record, so the constructed HDD bridge is part of the consumed Theorem 3.11 source data. A still stricter direct canonical-HDD wrapper now removes the preassembled `IUTStage1SourceHullDetObligations` input at this boundary: `ProductHullCanonicalHDDDirectCommonContainerSource` carries the product-hull, adjusted determinant, exact theta/family-hull calibration, q-positivity, and normalization data; Lean installs the record-canonical HDD bridge definitionally, transports the Theorem 3.11 record, and generates the hull/determinant obligations from the resulting constructor. The newest generated-full-label specialization, `GeneratedFullLabelProductHullDirectCanonicalHDDSource`, now ties this direct-HDD source to the generated Theorem 3.11 full-label quotient corridor: the multiradial record's possible images are the generated theta-class pullback, and the Step (xi) q-choice is the generated selected q-choice. The chosen-output generated full-label source, `ChosenOutputGeneratedFullLabelProductHullDirectCanonicalHDDSource`, lowers this once more by projecting that generated quotient corridor from the chosen-output primitive-packet route and then rebuilding the older generated-HDD source internally. The constructed-record version, `ConstructedRecordChosenOutputGeneratedFullLabelProductHullDirectCanonicalHDDSource`, first rebuilds the generated record's possible-image family from the chosen-output packet theorem and then lowers to the chosen-output source. The structured-bundle version, `StructuredGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, constructs the generated multiradial record from a generated SHE-structured bundle before entering the constructed-record source. The reindexed-structured version, `ReindexedStructuredGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, derives that generated bundle from the concrete structured bundle before entering the structured-bundle source. The constructed-qualitative version, `ConstructedQualitativeGeneratedChosenOutputFullLabelProductHullDirectCanonicalHDDSource`, derives that concrete structured bundle from constructed qualitative Theorem 3.11/SHE data and carries the qualitative and APT transport audits through the same reindexing. The corresponding preferred route, `preferredPublicConcreteStepXI311312ConstructedQualitativeGeneratedChosenOutputFullLabelProductHu` is definitionally the generated full-label route applied after these lowerings. These hardenings are now also available on one public surface: `preferredPublicConcreteStepXI311312RestrictionCalibratedGeneratedFullLabelProductHullDirectCanon` uses the generated full-label/direct canonical-HDD Step (xi) constructor while also requiring the restriction-calibrated local-global Frobenioid source for the p-adic Hodge local-prime comparison. Its audit pins both inputs: the generated full-label quotient corridor and direct-HDD constructor on the Step (xi) side, and the restriction-calibrated short-route audit on the p-adic side. This has now been tightened at the principal-product boundary: `ConstructedQualitativePrincipalProductHullOb3Ob5MeasureHodgeGapGeneratedChosenOutputFullLabelPro` requires the principal-product Ob3/Ob5 measure/Hodge-gap HDD source to be indexed by the constructed-qualitative generated record and synchronizes its q-choice with the selected generated q-choice. The corresponding restriction-calibrated route `preferredPublicConcreteStepXI311312RestrictionCalibratedConstructedQualitativePrincipalProductHu` enters the principal-product route through this coupled source. The direct-HDD audit now also exposes the restriction-calibrated goal-evidence package, so the inverse-dilation strict-selected branch remains visible after passing to the product-hull/HDD constructor layer. The separate current product-hull goal-evidence package records this non-vacuous principal-product route explicitly, avoiding confusion with the older quarantined norm-square compatibility audit. Its constructed-qualitative strengthening promotes the same evidence over the coupled constructed-qualitative/principal-product source, so this evidence surface no longer ranges over a generic constructor record at this point. A companion current evidence audit now couples that principal-product route to the restriction-calibrated norm-square/direct-summand projection feeding the Ob7 construction layer: Lean constructs the projected p-adic finite localized hull source, records the structure-sheaf/direct-summand synchronization and selected nonzero norm-square input, and exposes the restriction/Hodge-calibrated Ob7 source projection. This is a route-hardening step, not a completion marker, since the unit-ball norm-square branch remains diagnostic and quarantined from the inverse-dilation strict-selected branch. The same current evidence surface now has a structured-SHE version: Lean first constructs the principal-product direct-HDD source from the Remark 3.9.5 principal hull, Ob3/Ob4 adjusted determinant data, Hodge family-hull calibration, and structured Theorem 3.11/SHE side-condition package, so the source-obligation gap is no longer supplied as an assembled field at this current route boundary. A further adjusted-log-volume variant, `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeDirect` constructs that direct-HDD source from the record Ob3/Ob5 finite adjusted summand identity, deriving the normalized determinant equality internally instead of exposing it as a separate route input. The gap-backed variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeGapDir` lowers two more fields at this same boundary: q-pilot positivity and source normalization are projected from `IUTStage1SourceObligationGap`, while the adjusted Ob3/Ob5 source continues to derive the normalized determinant equality internally. The hodge-calibrated gap route `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeHodgeG` lowers the direct-HDD theta/family-hull equality as well: it is derived from the Hodge–Arakelov theta-monoid calibration of the bounded family-hull source canonically projected from the adjusted Ob3/Ob5 determinant source, together with the finite adjusted determinant endpoint. The measure-calibrated variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullOb3Ob5AdjustedLogVolumeMeasur` lowers the measure synchronization at the same boundary: Lean derives the Ob3/Ob5 hull-log-volume measure equality from the pre-ledger calibration to the product-hull holomorphic hull operator and the Ob3/Ob5/product-hull operator identification. The product-hull-constructed variant `preferredPublicConcreteStepXI311312RestrictionCalibratedProductHullConstructedOb3Ob5AdjustedLogV` lowers that operator-identification boundary as well: the Ob3/Ob5 adjusted source is built with the product-hull holomorphic hull operator, so the Ob3/Ob5/product-hull operator match is definitional rather than supplied as a separate compatibility field. The principal-product variant `preferredPublicConcreteStepXI311312RestrictionCalibratedPrincipalProductHullConstructedOb3Ob5Adj` lowers the product-hull source itself one level: the source carries the Remark 3.9.5 principal presentation by scalar multiples of the local integer region, and Lean projects the ordinary product-hull system internally. Below that public route, the constructor `principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallSource` now derives the Ob3/Ob4 source and adjusted-summand equality from the theta-region-defined principal valuation-ball cover. Its structured-SHE variant `principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallStructuredS` also derives the source-obligation gap from the structured Theorem 3.11/SHE bundle and side conditions. Its summand-charted Hodge variant `principalProductHullConstructedOb3Ob5SourceOfThetaRegionDefinedPrincipalValuationBallStructuredS` lowers the Hodge-family calibration one step further: the supplied Hodge input is the charted equality between the Hodge theta-monoid degree and the finite adjusted determinant summand sum, from which Lean derives the determinant, normalized determinant, and family-hull equalities internally. The remaining measure input is now represented by `IUTStage1PreLedgerHolomorphicHullMeasureCalibration`, a pointwise calibration saying that every target-region volume in the pre-ledger is the Remark 3.9.5 holomorphic-hull log-volume for the principal hull operator; Lean derives the equality of region-measure records consumed by the older route. The current product-hull/Ob7 diagnostic surface now has the corresponding theta-region constructor: Lean builds the principal-product HDD source from the theta-region principal valuation-ball data before assembling the current norm-square/Ob7 audit fields. That current audit also exposes the constructor-built Step (xi) boundary and the localized finite-place C_Θ endpoint/handoff supplied by the localized Step (xi) source; the structured-SHE and theta-region wrappers now project these same fields at their own concrete boundaries. The primitive wrapper `primitiveThetaRegionCurrentProductHullPrincipalHDDSource` removes the constructed-SHE bundle and transport guard from this HDD boundary by projecting them from the primitive Theorem 3.11 source constructor. The preferred public route now has the parallel theta-region route audit, and `preferredPublicConcreteStepXI311312PrimitiveTheorem311ConstructedThetaRegionGoalCompletionAudit` ties that route audit to the current norm-square/Ob7 evidence from the same theta-region source data while deriving the multiradial record, constructed-SHE bundle, and transport guard from the primitive Theorem 3.11 source constructor. The concrete packet route now builds this primitive source constructor through `IUTStage1PrimitiveTheorem311BridgeCertificateSourceConstructor`, which derives the primitive qualitative source from the transported IPL/SHE/APT bridge-certificate source. On the qualitative side, `IUTStage1SourceObligationGap` can now be constructed from a strengthened structured-SHE Theorem 3.11 bundle together with the q-positivity and normalization side conditions: the promoted SHE-alignment field is read from the structured SHE/common-container context rather than supplied as an independent equality, and the resulting obligations agree with the existing structured-input obligation constructor. The remaining boundary is lower: the original common-container/SHE and log-Kummer compatibility data still have to be derived from primitive paper-side Hodge-theater and log-theta-lattice constructions. The older norm-square public-surface and goal-scope evidence audits remain checked compatibility evidence, but now explicitly quarantine the p-adic unit-ball nonzero-norm boundary as well as the compact-open strict-positive boundary; they are not completion markers. The packet-facing goal-evidence refinement now projects the Theorem 3.11 source data from the concrete primitive packet and the Remark 3.9.5 principal hull from the packet/symmetry-label theta-region valuation-ball source before entering that norm-square evidence audit, so those two objects are no longer independent public inputs at this final compatibility boundary. A further restriction-calibrated packet refinement now projects the intrinsic finite-extension valuation cover and localized norm-square/direct-summand calibration from the pointwise restriction-calibrated local arithmetic source before entering the packet-facing final evidence audit; this branch remains diagnostic, since it also carries the unit-ball norm-square contradiction. The localized-C_Θ packet refinement now removes the standalone IUT IV C_Θ ledger, additive-Haar arithmetic-degree obligations, and remaining-payload witness from this same packet boundary: Lean projects them from the constructor-built localized Step (xi) local-term source before calling the restriction-calibrated packet evidence. The pointwise-localized packet refinement lowers this once more by constructing that localized C_Θ source from the pointwise localized Step (xi) determinant source, local main-log terms, Haar normalization defects, and finite-place summation identities before deriving the signed comparison and dichotomy at the packet-facing evidence layer. The arithmetic-divisor-localized packet refinement lowers that boundary again: the pointwise localized Step (xi) source is projected from the IUT IV arithmetic-divisor local evaluation ledger, whose local arithmetic-upper terms are identified with the Step (xi) weighted log-volumes, Haar defects, and main-log contributions before the same packet evidence is derived. The Theorem 1.10 valuation-ball packet refinement now lowers the IUT IV input one step further, projecting that arithmetic-divisor localized source from the valuation-ball additive-Haar formula-gap source before the packet evidence enters the arithmetic-divisor layer. The local-analytic valuation-ball packet refinement lowers the source once more: the formula-gap matched Theorem 1.10 source is projected from the local-analytic valuation-ball arithmetic-divisor source at the packet-facing boundary. The synchronized local-analytic packet refinement then constructs this local-analytic source from the constructor-built Step (xi) determinant/scale synchronization, the local-analytic arithmetic-divisor ledger, and the finite Haar/main-log comparison data. Its arithmetic-realized synchronized variant lowers this last comparison: it takes the explicit Theorem 1.10 local identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ and derives the older Step (xi)/Haar/main-log arithmetic-upper equality by unfolding the local arithmetic contribution. The formula-gap synchronized packet refinement lowers the local-analytic arithmetic-divisor input further by projecting it from the named valuation-ball formula-gap source carrying the Proposition 1.4 log-shell and Proposition 1.5 metric comparison records. Its arithmetic-realized companion performs the same recovery of the Step (xi)/Haar/main-log equality from $a_v(D_v + C_v) + E_v - M_v$ after projecting the local-analytic ledger from the formula-gap source. Across the pointwise localized, IUT-IV arithmetic-divisor, Theorem 1.10 valuation-ball, local-analytic, synchronized, and formula-gap packet audits, the constructed finite-place local-global C_Θ endpoint is now exposed explicitly rather than being visible only at the lower localized handoff. The pointwise Theorem 1.10 valuation-ball packet route now consumes the corresponding pointwise localized source, so the determinant/scale synchronization, formula-gap source, Haar defect, Step (xi)/Haar/main-log identity, and finite Haar lower bound are projected internally at this layer. This route is now a named audit record in its own right: it records the pointwise Theorem 1.10 endpoint, the internally constructed determinant/scale synchronization endpoint, the projected formula-gap packet evidence, and the resulting signed C_Θ comparison. The localized local-term packet route constructs that pointwise source from the localized Step (xi) local-term C_Θ source and the valuation-ball formula-gap source. A compatible refinement packages the remaining local arithmetic/main-log matching as a named source layer: the valuation-ball Theorem 1.10 local main-log function is the localized Step (xi) main-log summand,

and the local arithmetic upper contribution is the adjusted determinant log-volume plus Haar defect plus that summand. This replaces two anonymous equality families at the packet route by an audited compatibility source. The next arithmetic-realized refinement derives this compatibility from the expanded local Theorem 1.10 arithmetic-degree expression $a_v(D_v + C_v) + E_v - M_v$: the localized Step (xi) adjusted determinant plus Haar defect is matched with this local arithmetic-minus-main-log quantity, and Lean recovers the local arithmetic-upper equality by unfolding the Theorem 1.10 local contribution before entering the formula-gap synchronized packet audit. Lean now records this as a named arithmetic-realized packet audit, projecting the compatible packet evidence together with the localized C_Θ source, the local-term/IUT IV handoff audit, the signed C_Θ bound, and the Corollary 3.12 dichotomy directly. The formula-realized refinement goes one layer lower: it consumes the pointwise Hodge measure/summand formula source, constructs the theta-region and pointwise calibration records internally, and then enters the same arithmetic-realized packet audit. The local-arithmetic-degree refinement removes the packaged arithmetic-realization record at this boundary: it constructs that source from the main-log matching and the pointwise identity $\mathrm{StepXI}_v + \mathrm{Haar}_v = a_v(D_v + C_v) + E_v - M_v$ before recovering the localized C_Θ handoff and signed endpoint. The synchronized local-arithmetic-degree refinement goes one level lower again: from determinant/scale synchronization, the valuation-ball formula-gap ledger, the Haar defect, and the same local degree identity, Lean constructs the IUT IV arithmetic-divisor localized source and the localized local-term C_Θ source internally. Its constructed-synchronization companion now builds that synchronization record from the localized Step (xi) determinant source together with the constructor-built determinant and canonical-scale comparison equalities before entering this route. The norm-square localized determinant refinement projects this localized source from direct-summand norm-square vector-bundle data before using the same constructed-synchronization route. The p-adic finite-extension refinement now projects that norm-square determinant from unit-ball valuation-Haar compact-open local data before entering the same endpoint, and its synchronized local-upper variant packages the p-adic finite determinant together with the constructor determinant and canonical-scale sum identifications as one source object. The formula-gap local-upper variant now projects the local-analytic arithmetic-divisor ledger from the Theorem 1.10 valuation-ball formula-gap source before applying the same residual calculation. This boundary is now also represented by a typed residual-Haar source: the p-adic finite determinant/scale synchronization, the formula-gap source, and the total residual Haar lower bound are packaged together, and Lean projects this package to the localized Step (xi) C_Θ source, the local-to-global C_Θ handoff audit, the signed C_Θ inequality, and the dichotomy. The corresponding concrete compact-open restriction-calibrated public route now consumes this residual-Haar source directly, rather than taking the p-adic synchronization, formula-gap ledger, and residual inequality as separate final arguments. A further local-arithmetic handoff wrapper also consumes the product-level Ob7 handoff source, so the restriction-calibrated source, structure-sheaf/direct-summand equality, and nonzero summand norm are projected from one typed local-arithmetic package. The latest wrapper also consumes the pointwise measure/summand formula source, replacing the three public Hodge/measure formula equalities by a single source object. The current strongest compact-open wrapper consumes the named local Kummer realization map source, including its possible-region section and compact-open soundness laws. Lean forms the correspondence as the graph of this map, extracts the image laws, and then derives Remark 3.9.5 possible-region exactness, theta-region log-shell image laws, the local chart, and realized-exact compact-open sources internally. A pointwise Theorem 1.10 refinement also projects the determinant/scale synchronization and the arithmetic-degree Step (xi)+Haar identity from a single localized valuation-ball source. The construction of the arithmetic realization from foundational Hodge-theater/Frobenioid data remains a later boundary. Older public-looking route names remain implementation and compatibility layers below this citation point. The residue-module/unit-ball normalized restriction-calibrated source now also projects to the p-adic finite norm-square/direct-summand calibration source: Lean constructs the projected p-adic finite localized hull decomposition and then builds the norm-square source from the same restriction package. The remaining explicit inputs at this handoff are exactly the projected structure-sheaf/direct-summand synchronization on the p-adic unit-ball branch and the selected nonzero norm-square summand. The follow-up audit proves that this selected p-adic unit-ball norm is actually zero under the current normalization, so the handoff is diagnostic rather than non-vacuous; the additive-Haar inverse-dilation branch is therefore the side on which a non-unit selected factor still has to be constructed. That construction has now been exposed at the residue restriction-calibrated boundary: Lean projects the same residue-module source to the unified inverse-dilation localized-adjusted source and audits the selected inverse-base-prime additive-Haar factor. The audit proves strict Haar mass, positive normalized compact-open log-volume, and nonzero direct-summand norm for this selected factor, without reusing the quarantined p-adic unit-ball norm-square branch. The restriction-calibrated public route audit now includes this inverse-dilation strict-selected audit directly. Finite summation derives the Ob3-1 inequalities comparing the structure-sheaf log-volume with the localized bundle log-volume; the subtraction formula derives adjusted raw local positivity; positive determinant weights derive localized adjusted summand positivity; finite summation gives localized family-hull positivity. The theta scalar at the strict boundary is the environment theta divisor log-volume by definition. The public route now constructs the older Hodge-summand, Gaussian/Hodge, Hodge-evaluation, and Hodge-scalar sources internally from the Hodge canonical-degree divisor construction, audited evaluation, and localized adjusted-sum calibration, then derives the environment-theta/localized adjusted-sum equality; Lean then derives equality with the family-hull log-volume from the localized Ob3/Ob5 identity. This derived equality implies the family-hull/global calibration via theta/ordinary realified Frobenioid compatibility, and the localized determinant/family-hull equality then gives the determinant/global equality needed for the Gaussian-global nonzero proof. The earlier structure-sheaf/direct-summand boundary has also been lowered at the finite-extension endpoint: instead of publicly supplying 'structure-sheaf equals direct summand', the newest compact-open factor route uses the valuation-unit-ball normalized log-volume identity for the distinguished structure-sheaf factor and derives the direct-summand equality. The non-vacuous Haar-modulus/additive-Haar branch now has the same normalized-Haar-to-direct-summand handoff without passing through the vacuous unit-ball endpoint. On the residue-module/unit-ball normalized branch, the Hodge local-prime calibration has also been lowered: Lean now constructs the older normalized source from an environment-anchored local-global Frobenioid restriction source and a single ordinary-log-volume/Hodge-degree equality. Its preferred boundary audit also exposes the Step (xi-a)–(xi-g)/Ob1–Ob9 paper trace constructed from the same additive-Haar compact-open factor source, including the Ob7–Ob9 log-Kummer/vertical-shift/realified-log-volume entries. For strict positivity, the additive-Haar route now has a genuine Haar-mass-to-norm constructor, and the inverse-dilation finite-extension branch constructs a non-unit selected factor with strict Haar mass. The unit-ball finite-extension route remains as an explicit vacuity diagnostic. The earlier structure-sheaf-positive, raw-positive, summand-positive, explicit-ratio, weighted-partition, and one-summand localized constructors remain as compatibility layers. This is progress below a free-floating obligation bundle. What remains is to construct, from the source papers, the actual lower operations corresponding to:

the analytic construction of the class-indexed possible-image source,
the analytic construction of the place-audited packet/action symmetry-label transport inputs,
the analytic construction of the global-to-local restrictions and local-prime global equality,
the analytic construction of non-diagonal prime-place coordinates and the coric character,
the Hodge canonical-degree local-prime calibration, direct-summand/square determinant presentation, and point
the construction of the remaining theta-region local-field Haar factors,
and their calibration from the Hodge-theater packet data.

The strict route now derives the unit-ball tensor factors and one non-unit inverse-dilation strict factor from finite-extension local-field Haar data, and can build the record Ob3/Ob5 determinant/log-volume source from a principal-product theta-region cover identity and its finite additivity formula. The compact-open valuation-ball handoff now projects the final determinant identification from that record source at the exact possible-region, map, image, and correspondence boundaries. The exact-theta source-spine handoff can also construct its record Ob3/Ob5 determinant source from localized hull/vector-bundle data once the localized family is identified with the source-spine possible-image family; this now extends through tensor-label, label-transport, and symmetry-label transport, as well as the concrete target-region lattice formula and the concrete lattice-image law. The remaining work is to construct the full analytic valuation-cover family, its global bridge calibrations, and the Hodge-theater packet source that selects the appropriate non-unit local factors.

5.3 IUT IV estimates

The C_Θ comparison depends on IUT IV local and global estimates. The current formalization organizes the route through additive-Haar arithmetic-degree and p-adic formula objects, but the deepest estimates are still source obligations. What is now checked for the preferred route is the ordered-real and finite-sum handoff from the arithmetic gap to the canonical C_Θ-scale comparison, together with a constructor-built source-level bridge for the local-to-global handoff. In the older bridge, the constructed Step (xi) canonical scale is identified with the finite-place IUT IV scale before plus-one cancellation is applied. In the newer coupled bridge, the finite-place IUT IV scale is defined from the constructor-built Step (xi) canonical scale itself, so the exposed source assumption is the arithmetic gap inequality rather than a separate scale-identification equality. The finite-place constructor lowers this further: the arithmetic gap is derived from finite local Step (xi) summation data, namely local canonical scales, arithmetic/log contributions, local Haar defects, the local inequalities, and a global defect lower bound. The localized local-term constructor then lowers the local inequalities themselves by taking each local canonical scale to be the weighted adjusted log-volume of the localized Ob3/Ob4 determinant source and each local upper term to be Step (xi) log-volume plus Haar defect plus main-log contribution. This constructor is now fed directly by the arithmetic-divisor- backed additive-Haar source, which projects the local analytic and Theorem 1.10 C_Θ records through the formula-gap arithmetic-divisor source rather than assembling them inside the final finite-divisor route or its top-level paper-trace audit. The pointwise concrete-packet route now has the same lowering: it projects the raw IUT IV arithmetic-divisor local evaluation ledger from a valuation-ball additive-Haar formula-gap source before building the localized Step (xi) handoff. The Theorem 1.10 valuation-ball local-analytic localized source now also exposes the resulting localized C_Θ handoff audit, signed C_Θ bound, and dichotomy as source-level projections; the current product-hull evidence audit consumes these projections directly, rather than accepting an independent localized C_Θ source at that surface. The Record-Ob3/Ob5 valuation-ball source now projects to the closed valuation-ball formula-gap source and carries a DOD audit showing that the local

Haar summation, arithmetic gap, signed C_Θ bound, and dichotomy are inherited without adding a raw signed comparison premise. The named-HDD compact-open boundary now projects this same valuation-ball DOD audit, making the closed C_Θ chain visible at the route-input boundary rather than only in the lower source namespace. The canonical-HDD compact-open route audit also records this DOD evidence next to the reconstructed synchronized route input, the internally constructed local-analytic Step (xi) C_Θ source and handoff audit, the derived canonical-scale bound, signed C_Θ inequality, source-bridge inequality, Corollary 3.12 dichotomy, and definitional equality to the compact-open handoff. Its Hodge-formula companion also projects the pointwise measure/summand calibration from the formula source before invoking this current product-hull handoff. The local-arithmetic companion constructs the restriction-calibrated record from the residue-module/unit-ball Haar-character source, the realified Frobenioid restriction package, and the structure-sheaf/direct-summand normalization identities before invoking the same handoff. Its constructed-Theorem 1.10 companion now assembles the local-analytic IUT IV localized source from the localized determinant, scale-synchronization, arithmetic-divisor, Haar-defect, and finite-defect data before invoking the same current-product-hull route. Its synchronization-source variant removes the separate localized determinant source and determinant/scale equalities from this public surface, replacing them by one constructor-built localized Step (xi) synchronization source. What remains to construct from first principles is this lower IUT IV source itself:

the distinguished log-shell inclusions and numerical bounds,
the nondistinguished zero log-volume contribution,
the archimedean metric containment,
the arithmetic divisor source for Theorem 1.10,
the distinguished and archimedean formula-to-gap comparison sources,
the local finite-place Step (xi) inequalities and Haar defect bound.

Until these estimates and the upstream source spine are constructed from the published source data, the corridor remains conditional on source-paper mathematical content.

5.4 Local arithmetic and p-adic analytic infrastructure

The implementation has moved below a purely formal real-number boundary, but it still lacks significant local arithmetic infrastructure. The remaining work includes finite extensions of $\mathbb{Q}_p$, residue fields, Haar-normalized unit balls, explicit local degrees, different and conductor terms, and the formula matching that connects those objects to the arithmetic-degree route. This is standard mathematical infrastructure in spirit, but it is not small work in Lean.

6 Blueprint and Apodeixis

6.1 Lean blueprint status

The repository contains a Lean blueprint under `blueprint/`. It has a compact prose structure: aim and scope, the source Stage 1 route, the formal interface, the route to Corollary 3.12, open source obligations, and later work. It also has a generated declaration index, `blueprint/lean_decls`, which currently lists thousands of Lean declarations.

This means the blueprint is not empty, but it is still intentionally slim. It is adequate as a navigational document for the present stage. It is not yet a complete dependency graph of the project in the strongest sense: many nodes are marked not ready, and the prose chapters do not yet expand every major Lean route into a polished theorem-by-theorem mathematical dependency tree.

The right standard for the blueprint is different from the standard for this paper. The blueprint should be a map: concise nodes, Lean links, readiness status, and dependencies. This paper should be an argument: what we are formalizing, why the type boundaries matter, what is proved, and what remains mathematically open.

6.2 Apodeixis integration plan

The project is intended to be compatible with Apodeixis as a collaboration and review layer. The practical goal is to expose the formalization in a way that supports external mathematical review:

$$\begin{array}{c}\text{source-paper claim}\\ \Downarrow\\ \text{blueprint node}\\ \Downarrow\\ \text{Lean declaration or source obligation}\\ \Downarrow\\ \text{audit status and review discussion.}\end{array}$$

For this to work, the Lean code must avoid opaque omnibus assumptions, the blueprint must keep source claims linked to declarations, and the paper must avoid becoming a changelog. The current audit records are designed with this future use in mind. They give Apodeixis a natural unit of review: a named mathematical obligation with a precise Lean consumer.

7 Methodological Lessons

7.1 Typed boundaries are productive

The main technical lesson so far is that typed boundaries are not bureaucracy. They are mathematical tests. Once ordered real-line copies, transports, finite-label quotients, hull regions, source certificates, and upper-semi relations are assigned different types, informal shortcuts become impossible unless they are backed by explicit constructors.

This is especially useful in the 3.11–3.12 corridor because the public dispute is about whether a comparison has lost information during transport between different mathematical histories. Lean cannot decide that philosophical issue by itself, but it can prevent a formal proof from hiding the relevant movement.

7.2 Conditional routes are still useful

A conditional theorem can be scientifically useful if it names its conditions precisely. The current Stage 1 route is not a proof of Corollary 3.12 from the published IUT papers, but it has lowered the ambiguity of the problem. The remaining obligations are no longer "somewhere in the comparison". They are organized into Theorem 3.11 source construction, Step (xi) hull/determinant construction, IUT IV local-to-global estimates, and local arithmetic infrastructure.

7.3 The paper must not be a ledger

The earlier draft recorded too many implementation increments. That made it long but not more informative. The standing policy for future updates should be:

> update the relevant section,
> replace stale claims,
> move only stable mathematical outcomes into the paper,
> leave declaration-level detail to Lean and the blueprint.

This paper should remain regular research-paper length. The blueprint and generated declaration index are the correct place for detailed navigation.

8 Roadmap

The next milestones should eliminate source-obligation families rather than adding more endpoint wrappers.

Milestone A: Step (xi) hull/determinant construction. Continue lowering the remaining pointwise Hodge-calibration and local arithmetic fields below the constructed Remark 3.9.5/Step (xi) hull and determinant sources. The public route now constructs the principal hull, theta-region source, determinant handoff, localized Step (xi) C_Θ source, and signed endpoint from structured pointwise source objects. The latest compact-open additive-Haar variant also bundles the constructor-built determinant and canonical-scale identifications with the localized Step (xi) determinant into a single synchronization source, rather than exposing those identifications as separate public equalities. The same compact-open boundary now also has a pointwise Theorem 1.10/IUT IV variant in which the synchronization, formula-gap valuation-ball source, Haar defect, local arithmetic identity, and finite Haar lower bound are projected from one source; the p-adic formula-gap residual source performs the analogous finite-extension handoff by constructing the localized C_Θ source and signed dichotomy from the synchronized p-adic determinant and formula-gap ledger, and the concrete public wrapper now consumes that bundled source directly; the newest wrapper also replaces the three local-arithmetic restriction witnesses by the single product-level Ob7 handoff source, and then replaces the three measure/summand formula witnesses by the pointwise formula source; the current log-Kummer map wrapper further derives the correspondence, image laws, possible-region exactness, log-shell image, local chart, and realized compact-open exact sources from the named local realization map; the Record-Ob3/Ob5 synchronized companion now projects that source from the packet's canonical HDD record on the compact-open additive-Haar branch and audits the signed C_Θ comparison/dichotomy at that boundary. The next work is to derive the remaining pointwise Hodge and local-field inputs from the underlying Hodge-theater/Frobenioid data. The lower local-arithmetic boundary has also been moved, at both the named-HDD and component surfaces, from a supplied arithmetic-degree-controlled source to a constructed source obtained from valuation-ball formula-gap/p-adic data plus the Proposition 1.4/1.5 comparison synchronization; the remaining matching equality is now explicitly the equality to this constructed formula source. The product-handoff route has then been lowered one more step: the comparison package may now be projected from a local degree-identification formula source recording the Step (v) distinguished tensor formula, the Step (vii) archimedean coarse formula, and the arithmetic-degree coefficient lower bound. The concrete Record-Ob3/Ob5 component route now also combines this local-degree projection with the controlled-local-arithmetic construction of the valuation-ball formula-matching package, so neither the assembled valuation-ball formula source nor the comparison package is a separate public input at that handoff. A further p-adic Haar-index variant constructs the controlled local-arithmetic source itself from the valuation-ball local analytic arithmetic-divisor evaluation, unit-ball Haar defect, Step (xi) arithmetic-degree calibration, and the distinguished/archimedean formula bounds. Its direct controlled-product handoff now avoids the local-degree comparison reconstruction and only requires the Record-Ob3/Ob5 component equality with the p-adic Haar constructed formula source. The Record-Ob3/Ob5 component source now also projects its local degree-identification formula source internally from the arithmetic-divisor/effective-different/conductor component chain and definitionally preserves the formula core used by the comparison package. A projected-local-degree p-adic Haar route uses this projection to remove the standalone local-degree formula source and matching proof from that public surface; the remaining synchronization is the Record-Ob3/Ob5 component formula-gap equality against the p-adic Haar constructed formula source and its reconstructed comparison handoff on the local-degree branch.

Milestone B: IUT IV local-to-global C_Θ. Continue lowering the arithmetic-divisor-backed finite-place source. The distinguished, nondistinguished, and archimedean local processional estimates now feed the local C_Θ handoff through formalized Proposition 1.4/1.5 inequality chains, and the distinguished one-unit term can now be routed through the arithmetic-minus-main residual rather than a stronger local procession formula margin. The remaining work is to construct that arithmetic residual from the underlying local-field/Haar data and push the result through the finite-place summation. At the principal-pointwise packet boundary, the localized IUT IV arithmetic-divisor handoff now has a p-adic unit-ball specialization: the Haar defect is read from the finite-place unit-ball index source, and the pointwise identity $U_v = \mathrm{StepXI}_v + \delta_v + M_v$ constructs both the older localized C_Θ source and the Step (xi) synchronization source. The preferred principal-pointwise public endpoint now consumes this p-adic source directly before projecting to the older arithmetic-divisor route. This removes one abstract defect function from that public layer. A further processional p-adic route now constructs the same p-adic handoff from the Theorem 1.10 valuation-ball local-analytic source and the local Step (xi)-to-arithmetic synchronization $\mathrm{StepXI}_v = U_v - M_v - \delta_v$, so the p-adic public route no longer needs an already-packaged arithmetic-divisor p-adic source when these processional data are available. Its synchronized variant now also reads the localized determinant source and the canonical C_Θ-scale finite-sum identity from the constructor-built/localized Step (xi) synchronization source, rather than exposing those two equalities separately. A further processional-estimate variant constructs the bundled p-adic processional synchronization from the local p-adic unit-ball Step (xi) identity and the Theorem 1.10 local-analytic processional-estimate source, with a single identification between the normalized p-adic place and the distinguished finite place. Thus this public route no longer takes the processional p-adic synchronization record as an independent source object. The normalized-place refinement removes that last identification as a public hypothesis: the processional estimate is stated directly at the p-adic unit-ball normalized place, so the distinguished nonarchimedean place is no longer chosen independently at this boundary. A further localized-local-term variant reads the same localized determinant and scale data directly from the already-derived localized local-term C_Θ source, leaving only the local-analytic processional Step (xi) identity at that boundary. The localized-local-term handoff now also has an equality-free p-adic-normalized constructor: ofLocalizedStepXILocalTermCThetaSourcePadicNormalizedProcessionalEstimateSource combines the local-term determinant/scale source, the p-adic Step (xi) identity, and processional-estimate data whose distinguished place is definitionally the p-adic normalized place. Thus the bundled processional p-adic synchronization source and the normalized-place equality are no longer separate inputs at that constructor surface. The newest pointwise Theorem 1.10 variant projects the localized determinant data and local-analytic valuation-ball ledger from the pointwise IUT IV localized source itself, so the p-adic processional endpoint no longer takes a separate localized local-term source at this layer. A new Haar-residual pointwise endpoint now rebuilds the bundled processional p-adic source from that pointwise source and the expanded p-adic unit-ball Haar-index arithmetic-degree residual identity before deriving the signed Corollary 3.12 dichotomy. The normalized pointwise endpoint remains the equality-free processional-estimate route: it uses the p-adic Step (xi) synchronization identity and p-adic-normalized processional estimates. The local-analytic pointwise refinement lowers this once more: the formula-gap valuation-ball package is no longer exposed at the p-adic public boundary; the endpoint reads the localized Step (xi) determinant object, determinant and scale comparisons, and Theorem 1.10 local-analytic ledger from the pointwise local-analytic localized source. Its Haar-residual public wrapper now removes the bundled processional p-adic synchronization from that endpoint surface by deriving it from the expanded residual source. A still lower public wrapper now separates that expanded residual source from the processional-estimate ledger, retaining only the equality that the distinguished processional place is the p-adic normalized place. The strongest current p-adic public wrapper combines this with the p-adic-normalized convention and then weakens the processional side once more: it exposes the pointwise local-analytic source, the expanded p-adic Haar-residual identity, and p-adic-normalized processional case data. The case source records the case-by-case normalized log-volume identities and the p-adic normalized-place kind, but no longer asks for the stronger distinguished $+1$ processional upper-bound field; the extra unit contribution is supplied by the Haar-residual lower bound. The endpoint therefore constructs the p-adic Step (xi) synchronization and distinguished alignment internally. This boundary has now been lowered once more: the p-adic unit-ball/local-analytic endpoint takes a p-adic unit-ball localized Step (xi) source, a matching valuation-ball local-analytic Theorem 1.10 ledger, their local arithmetic-evaluation equality, and the normalized processional case data. Lean projects the pointwise local-analytic localized source and derives the expanded p-adic Haar-residual arithmetic-degree source internally. The same reconstructed local-analytic handoff now also proves the $C_\Theta \geq -1$ companion at this p-adic unit-ball boundary, so the lower bound no longer has to pass through the older public local-analytic source argument. The lower-bound projection has now also been lifted through the stronger processional p-adic source, using its embedded valuation-ball local-analytic ledger and p-adic unit-ball projection. The remaining work is to derive this smaller source package from the underlying Theorem 3.11/Hodge-theater and local-field constructions.

Milestone C: Theorem 3.11 source derivation. Work backward from the typed indeterminacy core to actual Hodge-theater, Frobenioid, log-Kummer, and log-theta-lattice source data. This is the largest milestone and the one most directly connected to the Scholze–Stix objection.

Milestone D: Blueprint expansion. Expand the blueprint from a slim guide plus declaration index into a richer review graph. Each nontrivial source obligation should have a node with source citations, Lean declarations, readiness status, and a short explanation of the mathematical construction still missing.

9 Conclusion

The project has reached a useful but conditional Stage 1 state. Lean verifies the final real inequality, distinguishes the disputed comparison levels, keeps $\mathrm{Ind}_1/\mathrm{Ind}_2$ equality behavior separate from Ind_3

upper-semi behavior, and threads a concrete Hodge-theater/log-theta-style source route into the current C_Θ/Corollary 3.12 endpoint. The code is not merely a prose annotation of the papers; it is an executable audit of a precise corridor.

What remains is real mathematics, not formatting. To settle the 3.11–3.12 issue inside this project, the source obligations must be discharged from Mochizuki's constructions: full Theorem 3.11 multiradial data, genuine Remark 3.9.5 hull/determinant operations, IUT IV local estimates, and the local arithmetic infrastructure connecting them. The formalization has made that task sharper. It has not made it disappear.

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